

Viet Nam Import & Export

Implement the Model



Group 6



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Prophet Model

In this project, we use the **Prophet** model developed by Facebook (Meta)
Prophet is a time series forecasting model that works well with data that has trend and seasonal patterns.

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t$$

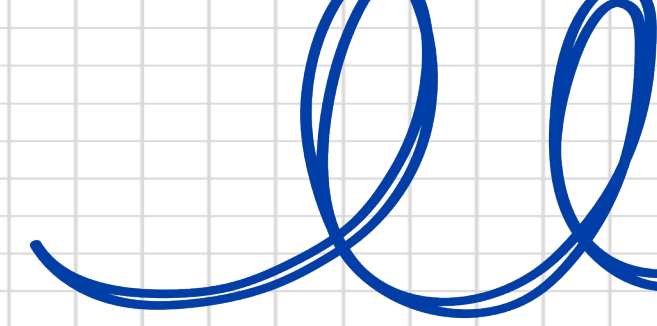
Where:

- **g(t)** represents the trend (long-term increase or decrease)
- **s(t)** represents the seasonality (repeating patterns such as months or years)
- **h(t)** can include holiday effects or special events
- **ε_t** is random noise





Comprehension about Trend Modeling - $g(t)$



Month	Export (VND)
1	100
2	120
3	130
4	150
5	170
6	190
7	210
8	240
9	260
10	280
11	300
12	320

From the data, we can see:

- From month 1 to 6: export increases slowly (+20 per month).
- From month 7 to 12: export grows faster (+30 per month).

1. Base Trend $g_0(t)$

Prophet first fits a simple straight line for all data:

$$g(t) = k.t + m$$

For example:

- $k = 20$ (growth per month)
- $m = 80$ (starting point)

2. Changepoint (t_j)

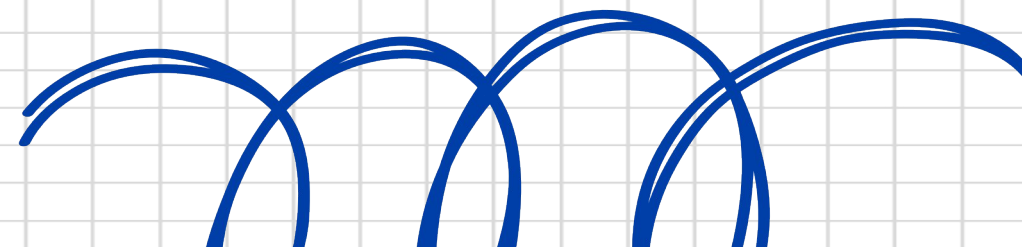
Prophet checks where the error increases and finds a changepoint, here at month 7.

After this point, Prophet allows the slope (k) to change:

$$K(new) = k + \delta_1$$

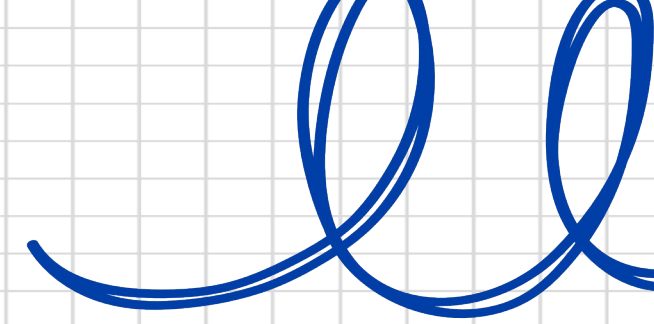
If $\delta_1 = +10$, then the new slope = $20 + 10 = 30$.

→ The trend becomes steeper after month 7.





Comprehension about Trend Modeling - $g(t)$



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From the data, we can see:

- From month 1 to 6: export increases slowly (+20 per month).
- From month 7 to 12: export grows faster (+30 per month).

3. Offset Correction (γ)

When the slope changes, the line can “jump” at the changepoint. Prophet adds γ_1 (gamma) to move the line up or down, keeping it smooth and continuous.

$$\gamma_1 = -\delta_1 \cdot t_1$$

With $\delta_1 = +10$ and $t_1 = 7 \rightarrow \gamma_1 = -70$.
So Prophet corrects the intercept by -70 to keep the two parts connected.

4. Final Trend Function

The complete trend is:

$$g(t) = (k + \alpha(t)\delta)t + (m + \alpha(t) \cdot \gamma)$$

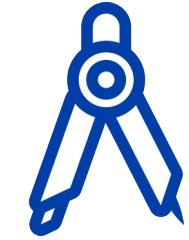
where $\alpha(t)$ is 0 before month 7 and 1 after month 7.

- For $t < 7$: $g(t)=20 \cdot t+80$
- For $t \geq 7$: $g(t)=(20+10) \cdot t+(80-70)=30 \cdot t+10$

Before month 7: slow growth (+20 per month)
After month 7: faster growth (+30 per month)
Both lines meet smoothly at month 7.



Comprehension about Seasonality Modeling – $s(t)$



t (month)	y (export value)
1	100
2	120
3	130
4	150
5	170
6	190
7	210
8	240
9	260
10	280
11	300
12	320

1. Build the formula

With period $P = 12$ and $N = 1$: (Fourier series)

$$y_t = a_1 \cos\left(\frac{2\pi t}{12}\right) + b_1 \sin\left(\frac{2\pi t}{12}\right) + \epsilon_t$$

Goal:

Find a_1 and b_1 that make the **Loss** (error) as small as possible:

$$L(a_1, b_1) = \sum_{t=1}^{12} (y_t - a_1 \cos(\frac{2\pi t}{12}) - b_1 \sin(\frac{2\pi t}{12}))^2$$

2. Matrix form

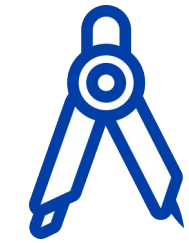
We can write this as a matrix equation:

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{12} \end{bmatrix}, \quad X = \begin{bmatrix} \cos(\frac{2\pi 1}{12}) & \sin(\frac{2\pi 1}{12}) \\ \cos(\frac{2\pi 2}{12}) & \sin(\frac{2\pi 2}{12}) \\ \vdots & \vdots \\ \cos(\frac{2\pi 12}{12}) & \sin(\frac{2\pi 12}{12}) \end{bmatrix}$$

Then the best coefficients are found by:

$$\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = (X^T X)^{-1} X^T y$$

Comprehension about Seasonality Modeling – $s(t)$



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3. Calculation result

Using linear regression:

$$a_1 = 25.39, b_1 = -77.75$$

So, the **seasonal function** is:

$$s(t) = 25.39 \cos\left(\frac{2\pi t}{12}\right) - 77.75 \sin\left(\frac{2\pi t}{12}\right)$$

4: Compute Loss (SSE)

$$L = \sum_{t=1}^{12} (y_t - s(t))^2 = 36,095.55$$

This is the **minimum error** Prophet can reach using the two parameters a_1 and b_1 .

If more harmonics are added ($N=2,3,\dots$), Prophet will have more parameters $\rightarrow$ more complex seasonal path $\rightarrow$ loss decreases.

N = number of seasonal waves (pairs of sine and cosine).

Comprehension about Holiday & Events – $h(t)$



In Prophet, $h(t)$ represents the **effect of holidays or special events** on the forecast.

$$h(t) = \sum_{i=1}^L \delta_i \cdot D_i(t)$$

Where:

L – number of holidays or events.

$D_i(t)$ – dummy variable:

+ $D_i(t) = 1$ if time t is within event i ,

+ $D_i(t) = 0$ otherwise.

δ_i – the estimated impact of event i on $y(t)$, learned by Bayesian regression.

+ If $\delta_i > 0 \rightarrow$ the event increases $y(t)$.

+ If $\delta_i < 0 \rightarrow$ the event decreases $y(t)$.

Example

Suppose there are 12 months with **two events**:

- **January:** New Year $\rightarrow$ increases by 20
- **August:** Trade Fair $\rightarrow$ increases by 15

Then:

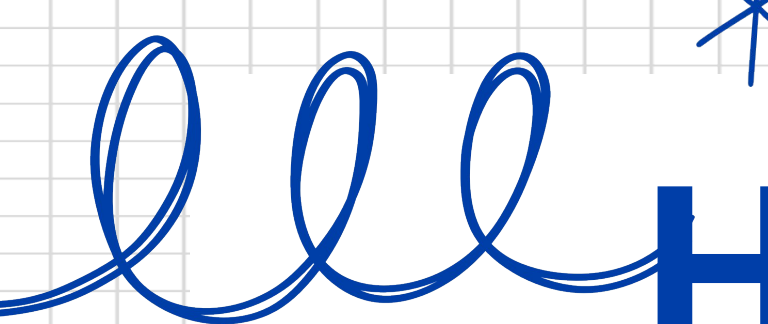
$$D_1(t) = \begin{cases} 1 & \text{if } t = 1 \\ 0 & \text{otherwise} \end{cases}, \quad D_2(t) = \begin{cases} 1 & \text{if } t = 8 \\ 0 & \text{otherwise} \end{cases}$$

With $\delta_1 = 20$ ε $h(t) = 20 \cdot D_1(t) + 15 \cdot D_2(t)$

When $t = 1 \rightarrow h(1) = 20$

When $t = 8 \rightarrow h(8) = 15$

For other months $\rightarrow h(t) = 0$



How The Model Works

Step 1: Data Preparation

Prophet needs two main columns:

- **ds**: the date or time column
- **y**: the value we want to predict (for example: export value)

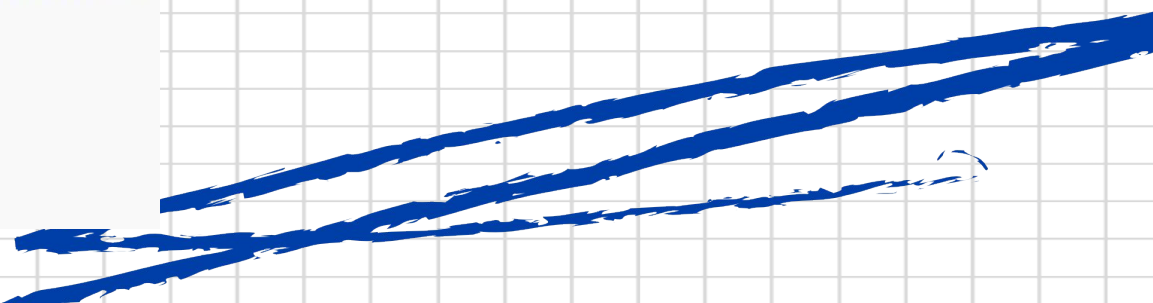
The data should be in time order. Prophet can still work even if some months are missing.

Step 2: Fit the Prophet Model

Create the model and fit it to your data:

This step helps Prophet learn the pattern and trend in your data.

```
m <- prophet(  
  growth = "linear",  
  yearly.seasonality = TRUE,  
  weekly.seasonality = FALSE,  
  daily.seasonality = FALSE,  
  changepoint.prior.scale = 0.5,  
  interval.width = 0.95  
)  
m <- fit.prophet(m, data)
```



How The Model Works

Step 3: Make Future Dataframe

Create a new time series for the next 12 months:

```
future <- make_future_dataframe(m, periods = 12, freq = "month")
```

Step 4: Forecast Future Values

Predict the export values for the next 12 months:

```
forecast <- predict(m, future)
```

Prophet gives three main results:

yhat: predicted value

yhat_lower: lower bound of forecast interval

yhat_upper: upper bound of forecast interval

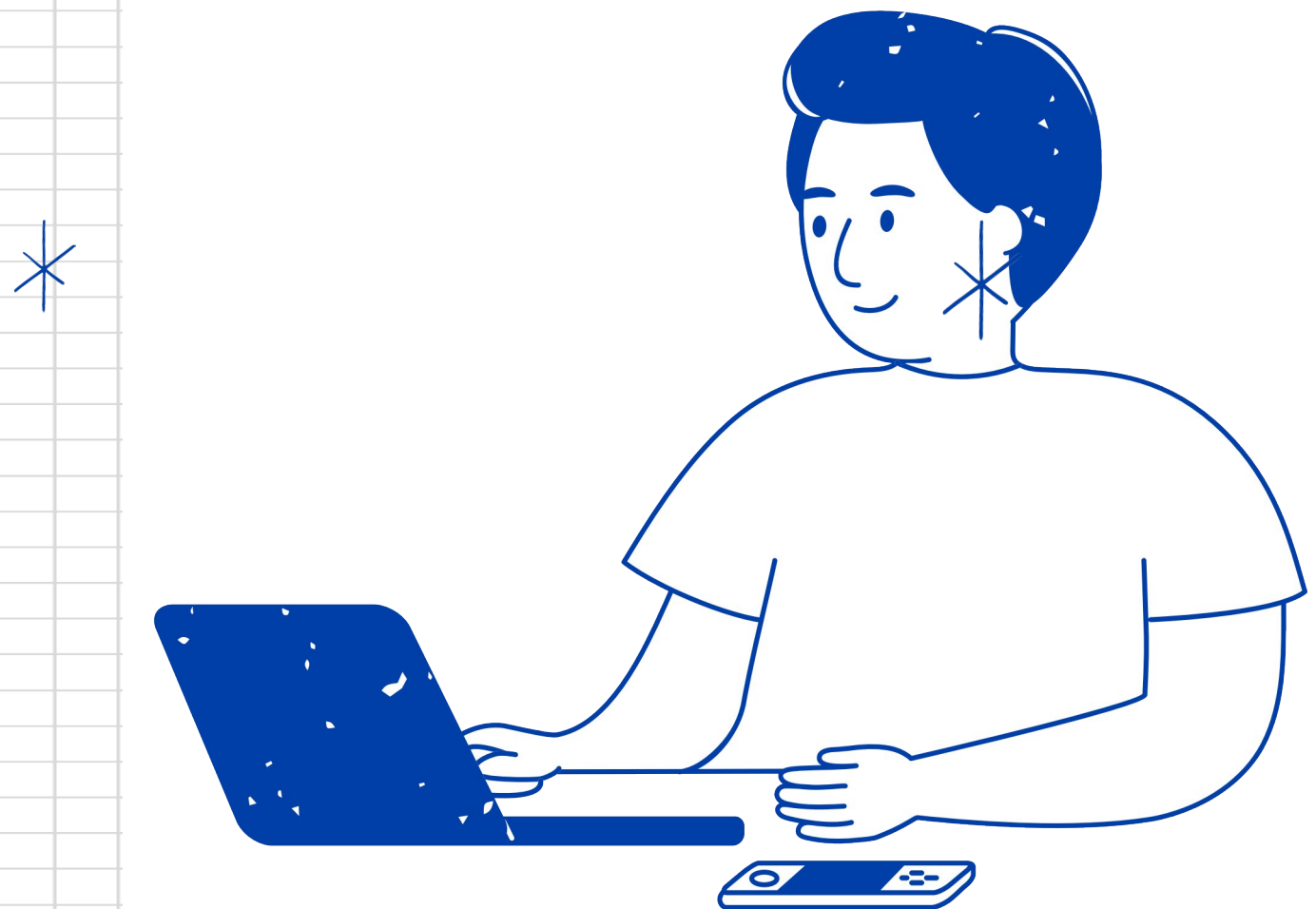
Step 5: Visualize the Forecast

The black dots show the actual data,
and the blue area shows the forecast with confidence interval.

Step 6: Evaluate the Model

Compare predicted values with real data.

Use metrics such as **MAPE** and **RMSE** to check accuracy.



Parameter of Model

1. Growth

- **Options:** “linear” or “logistic”.
- **Meaning:** Controls the overall trend Prophet uses.
 - **Linear:** The series increases or decreases steadily over time.
 - **Logistic:** Growth slows down when reaching a maximum value (saturation).

2. Changepoint.Prior.Scale

- **Default:** 0.05
- **Meaning:** Decides how much the trend can change at certain points.
- **Effect:**
 - Small value (e.g., 0.01) → smoother trend, less flexible.
 - Large value (e.g., 0.5) → more flexible, follows sudden changes.

3. Yearly.Seasonality

- **Default:** auto
- **Meaning:** Adds patterns that repeat every year.
- **Effect:** Helps model yearly cycles such as monthly export changes.
- **Tip:** Set **TRUE** when you have more than two years of data.

4. Weekly.Seasonality

- **Default:** auto
- **Meaning:** Adds patterns that repeat every week.
- **Use:** Only for **daily** data.
- **Tip:** Set **FALSE** for monthly or yearly data.

5. Daily.Seasonality

- **Default:** auto
- **Meaning:** Adds daily patterns (hourly cycles).
- **Use:** For hourly or minute data only.

6. Interval.Width

- **Default:** 0.80
- **Meaning:** The size of Prophet’s prediction interval (uncertainty range).
- **Effect:**
 - 0.95 → wider interval (more uncertainty).
 - 0.80 → narrower interval (less uncertainty).

7. Changepoint.Range

- **Default:** 0.8
- **Meaning:** The proportion of history where Prophet can add changepoints.
- **Effect:**
 - Small value → fewer trend changes.
 - Large value → more flexibility near the end.

8. Fourier.Order: 10

- **Meaning:** Number of sine and cosine pairs Prophet uses to model seasonality.
- **Effect:**
 - Small value (e.g. 3) → smooth, simple seasonality.
 - Large value (e.g. 15) → more complex seasonality, can overfit.

Metric of Model

1. MAPE (Mean Absolute Percentage Error)

→ shows the average percentage difference between the predicted and actual values.

→ lower is better.

$$MAPE = \frac{1}{n} \sum \left| \frac{actual - predicted}{actual} \right| \times 100$$

2. RMSE (Root Mean Squared Error)

→ measures how far predictions are from actual values.

→ lower RMSE means the model predicts closer to reality.

$$RMSE = \sqrt{\frac{1}{n} \sum (actual - predicted)^2}$$

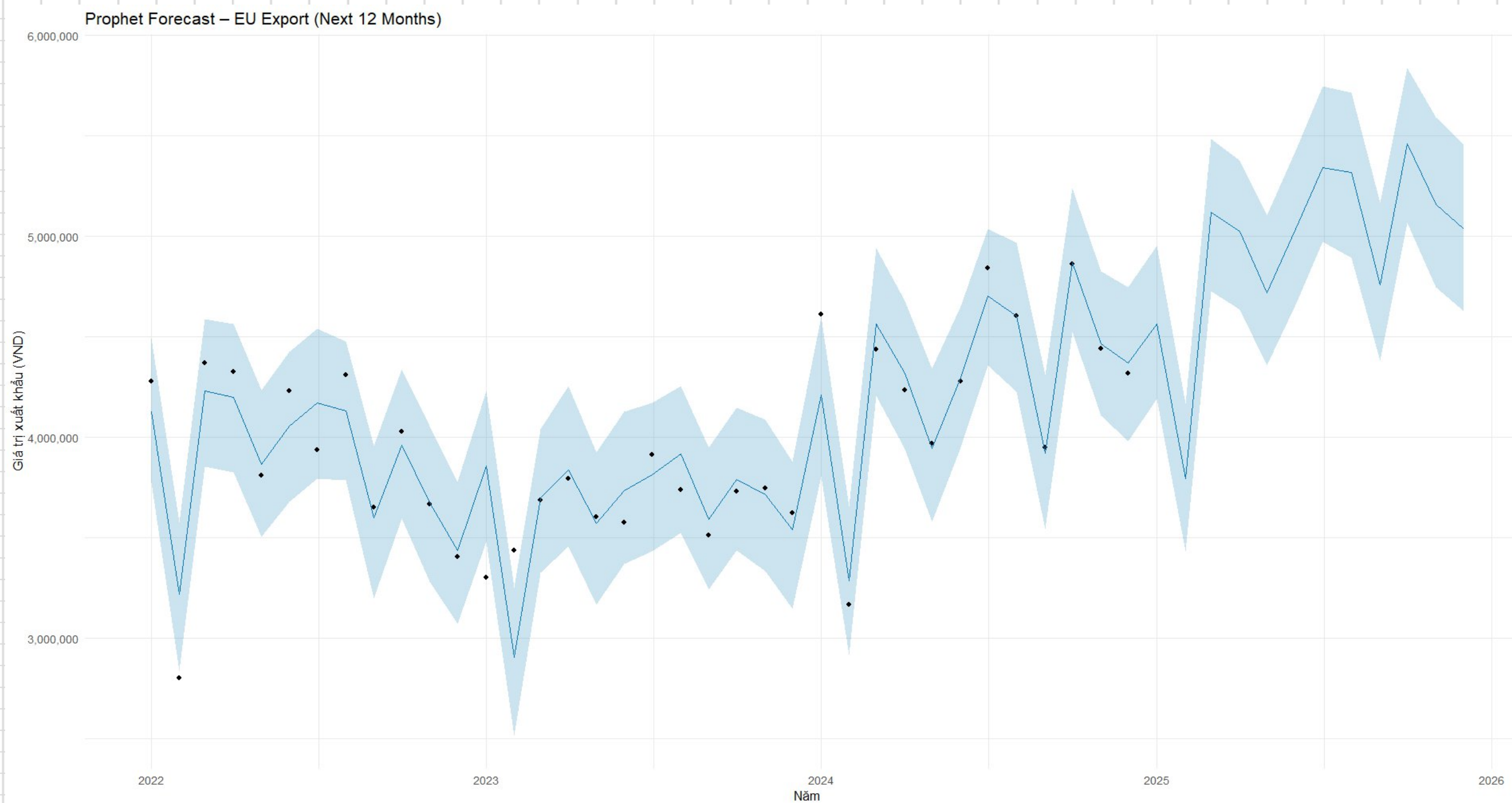
Result of Model

	EU	ASEAN	United States	Japan	Korea
MAPE	8.8 %	16.48 %	6.05 %	31.31 %	4.88 %
RMSE	126768.6	834600.5	101389.2	621258.3	295817.6

	Australia	China	India	Ukraine
MAPE	11.57 %	7.85 %	16.08 %	549.81 %
RMSE	76605.06	1195641	90225.52	57802.77

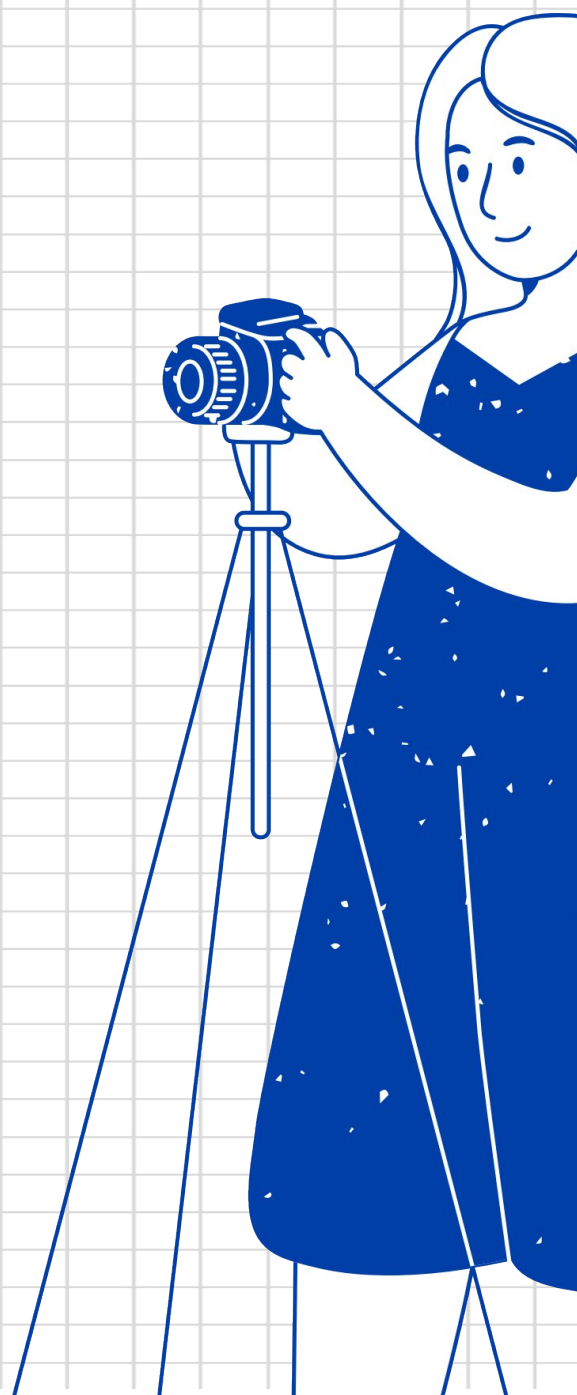
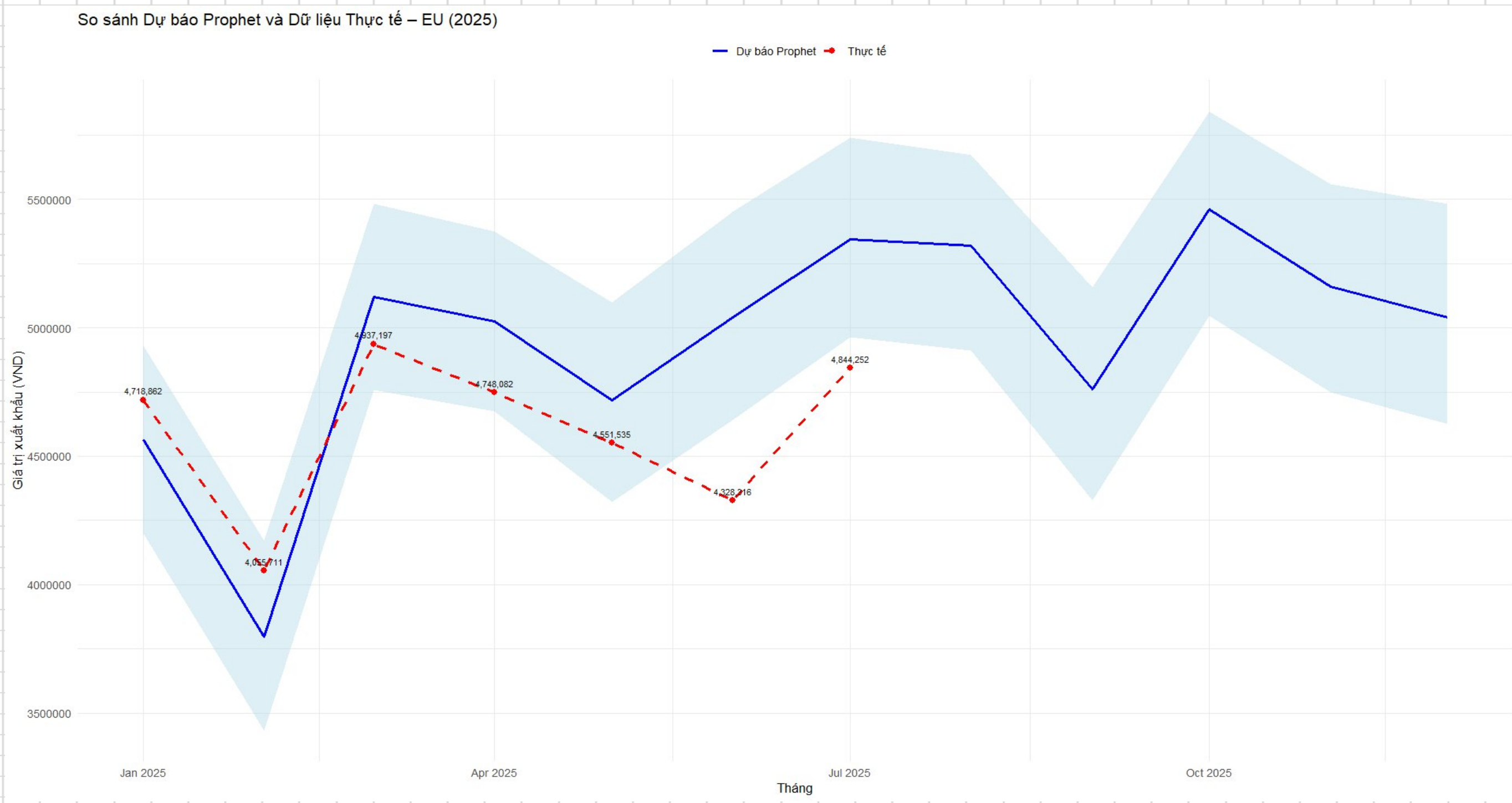
Implement Model

EU



Implement Model

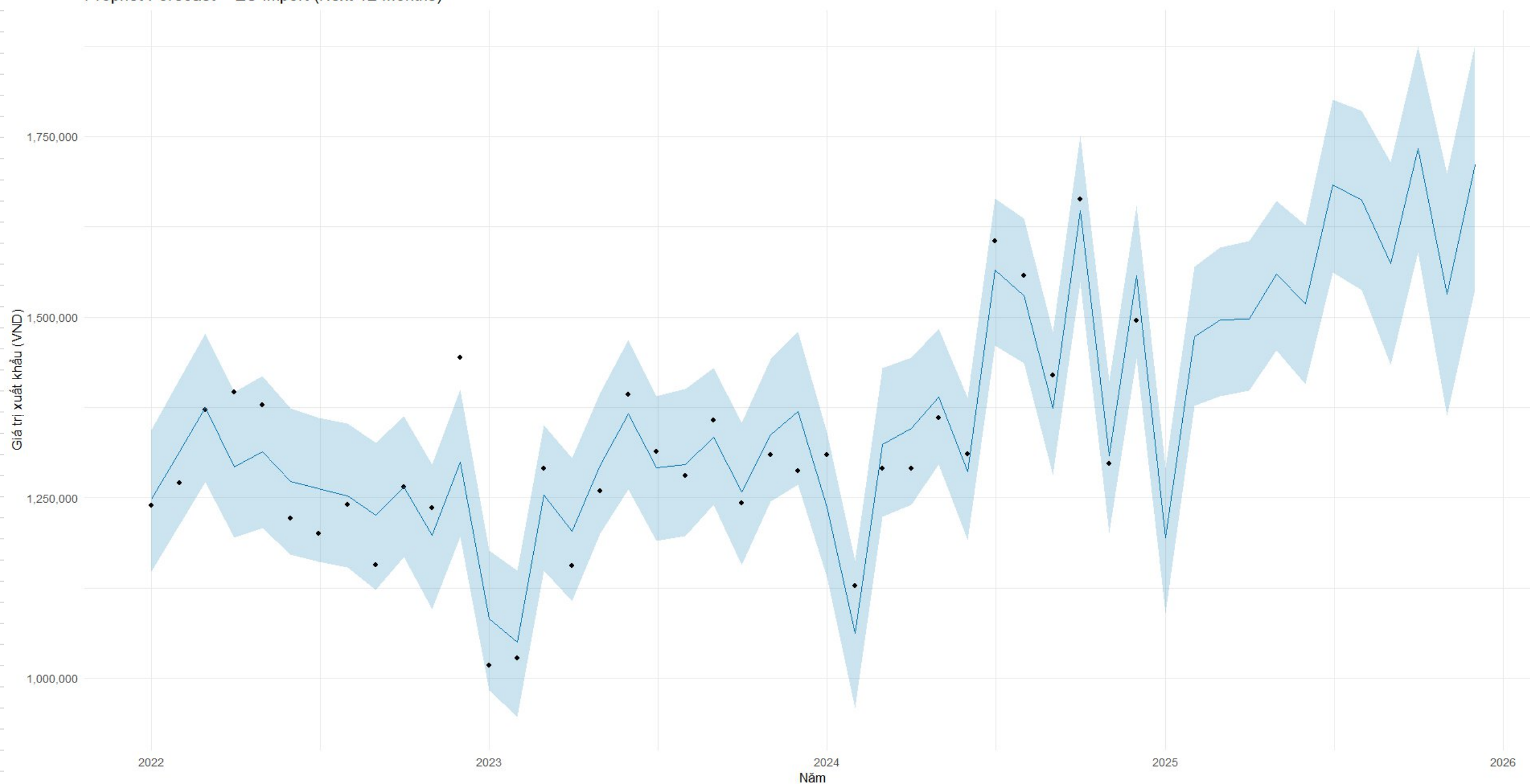
EU



Implement Model

EU

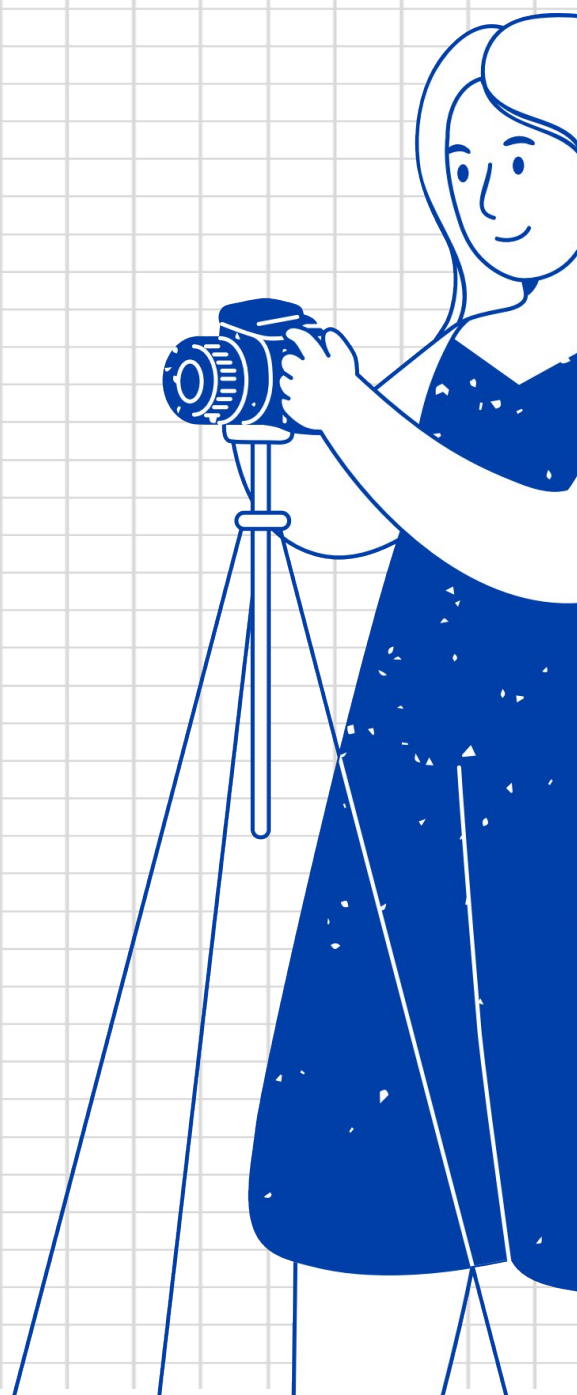
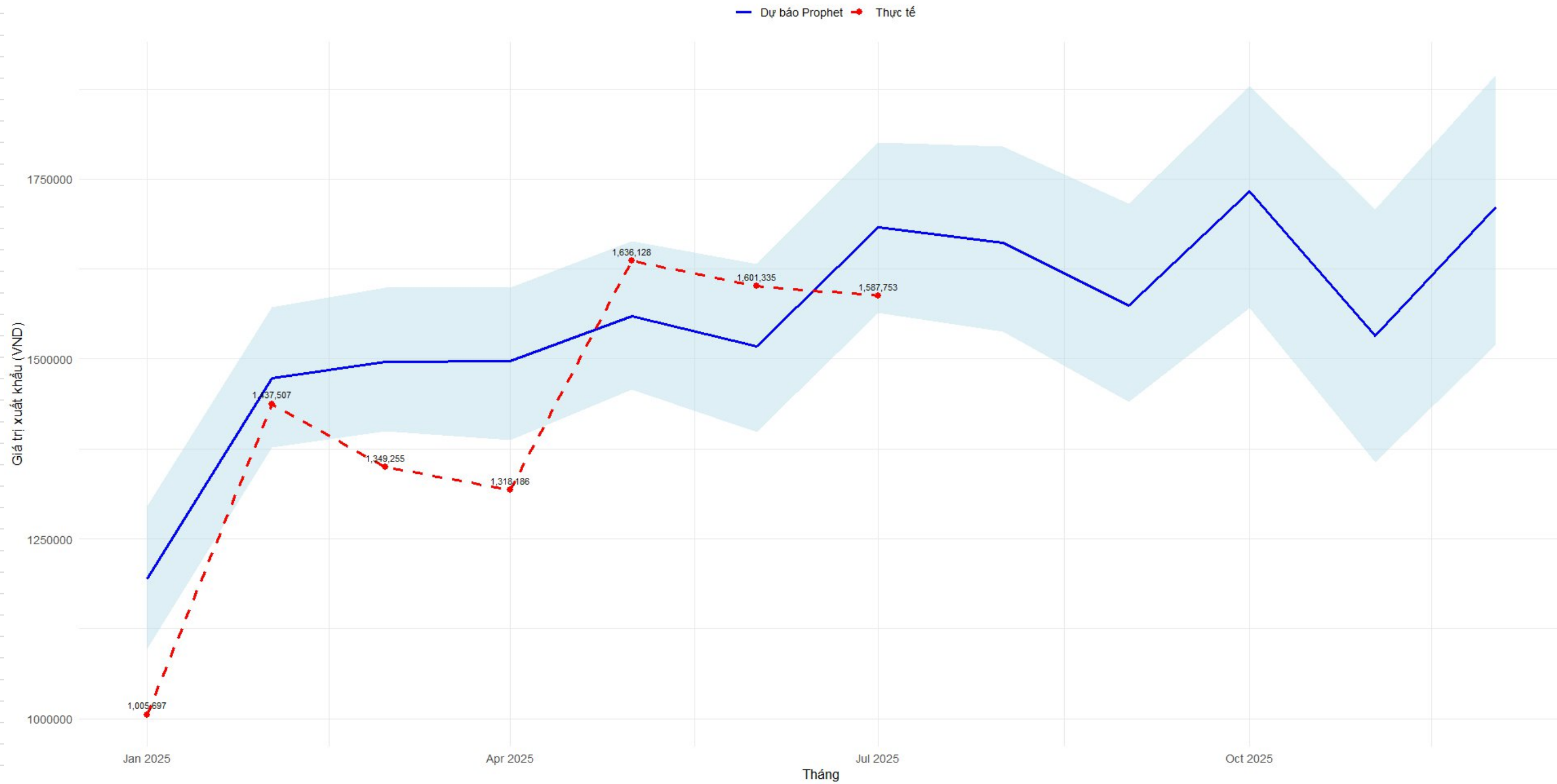
Prophet Forecast – EU Import (Next 12 Months)



Implement Model

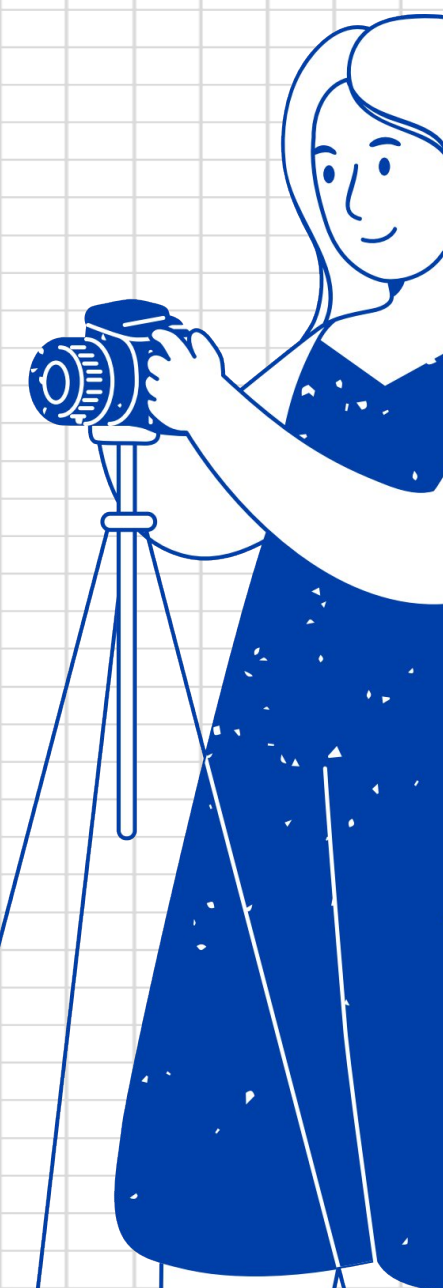
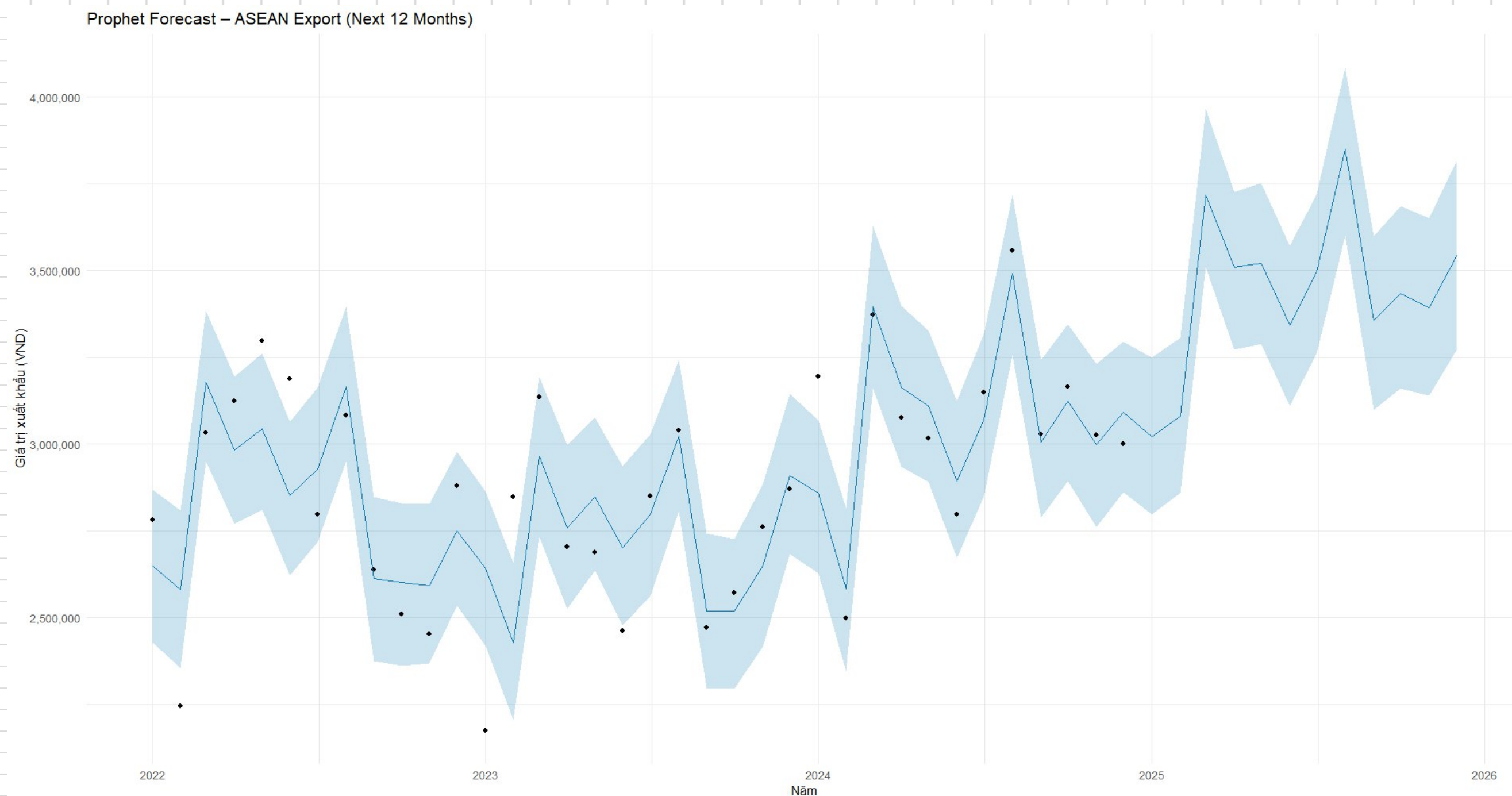
EU

So sánh Dự báo Prophet và Dữ liệu Thực tế – EU (2025)



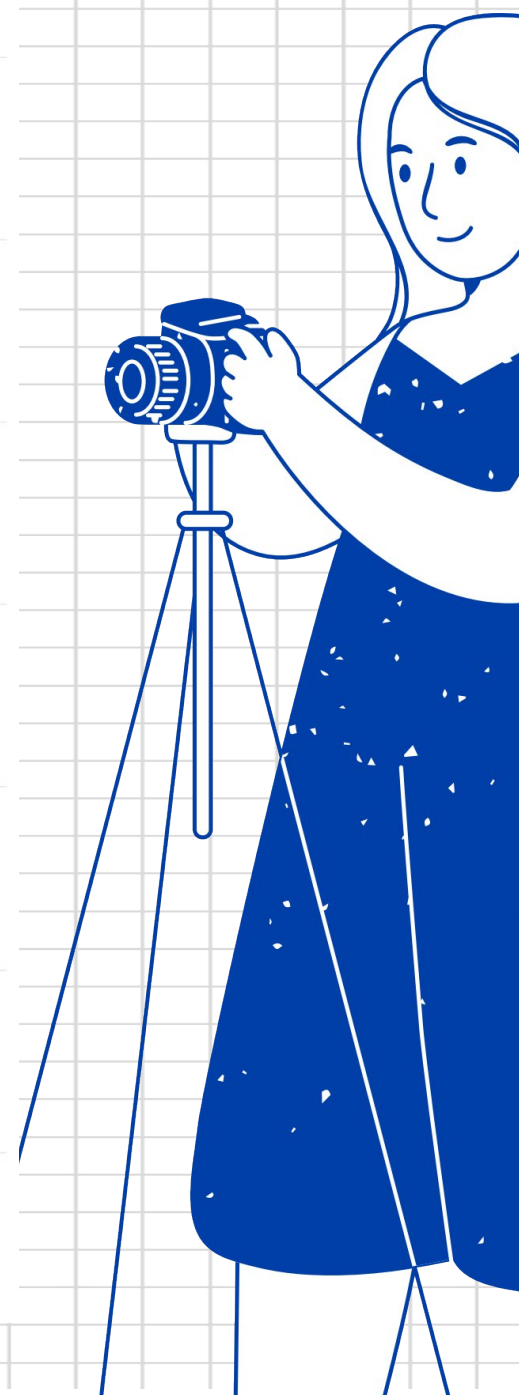
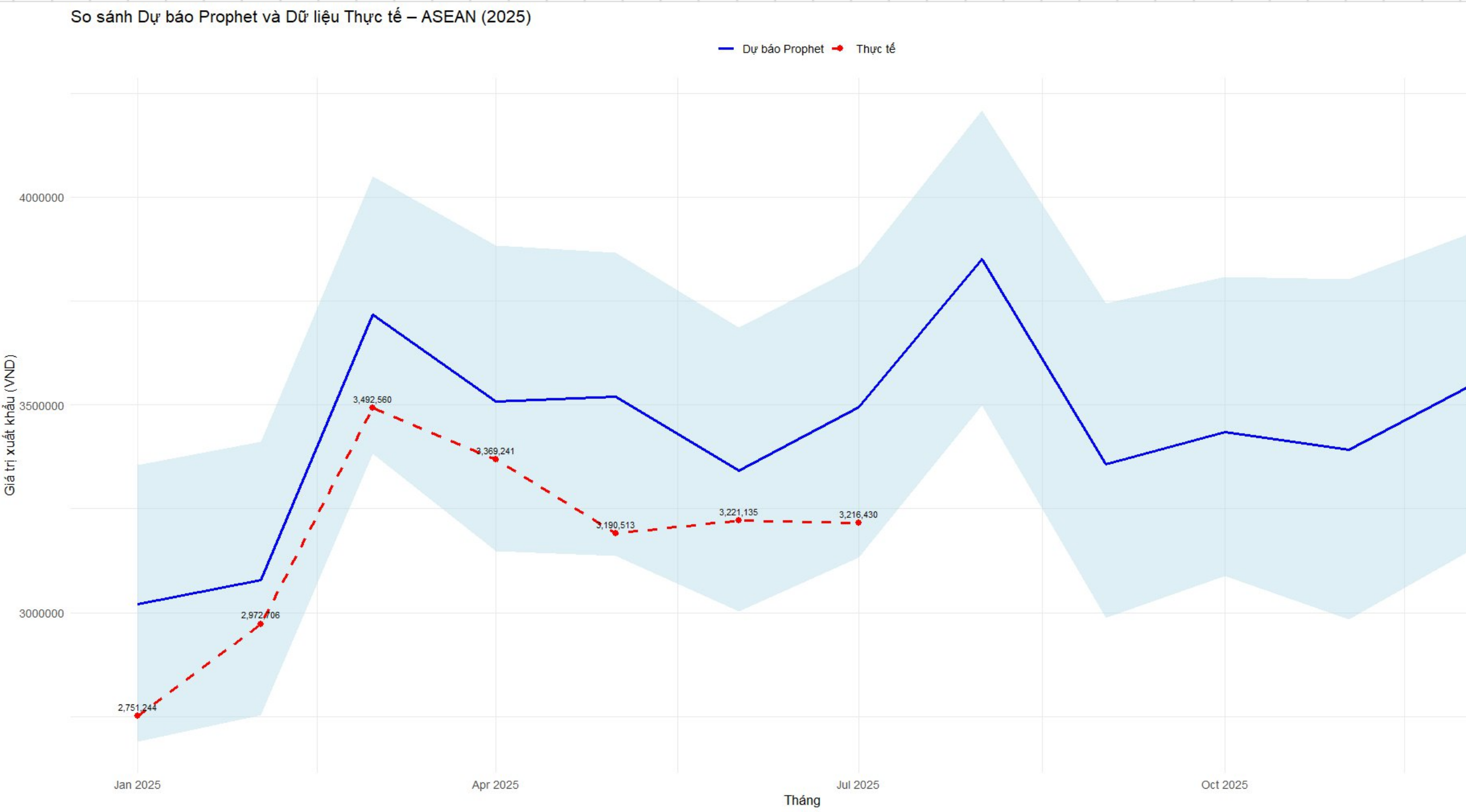
Implement Model

ASEAN



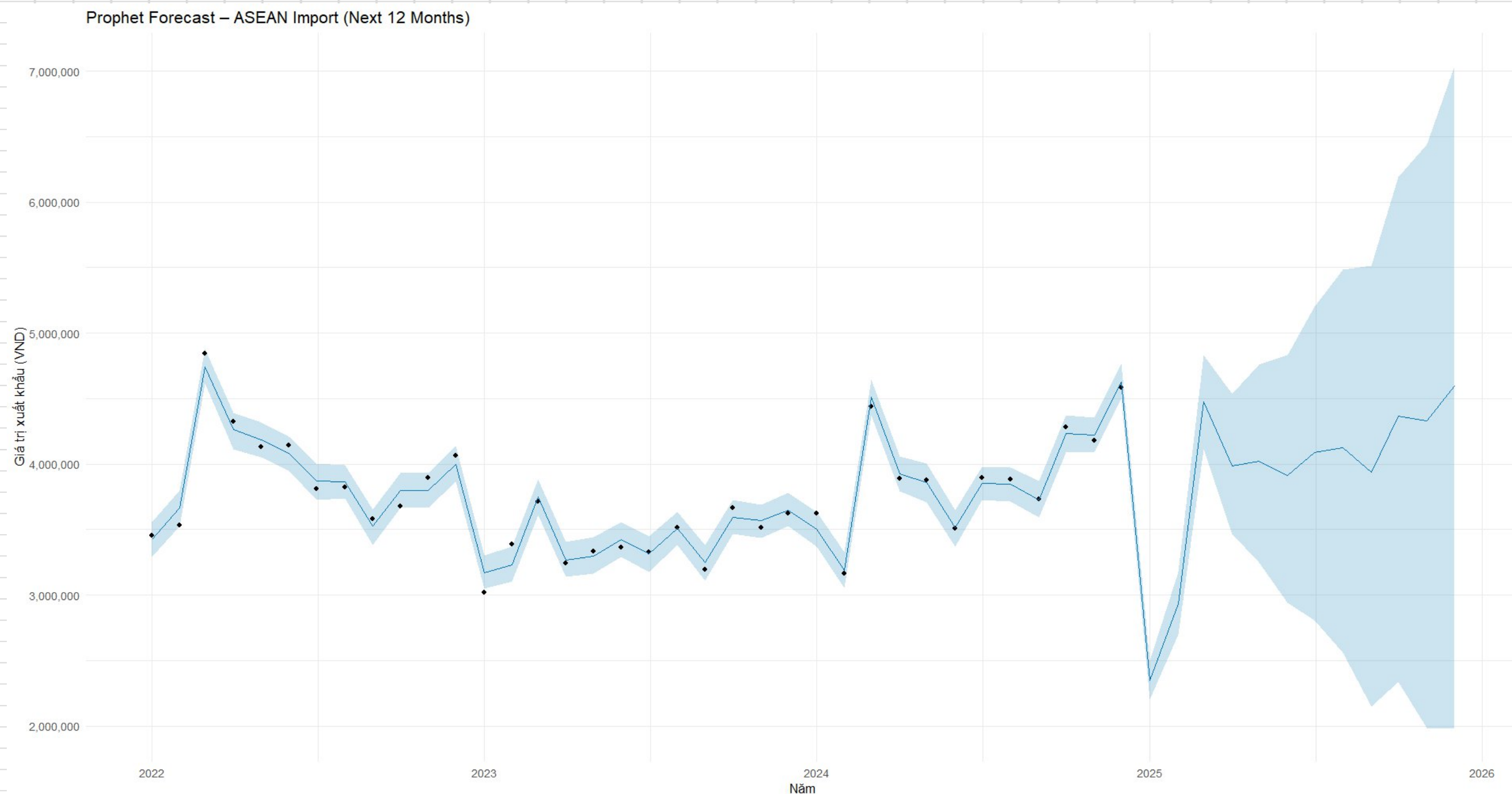
Implement Model

ASEAN



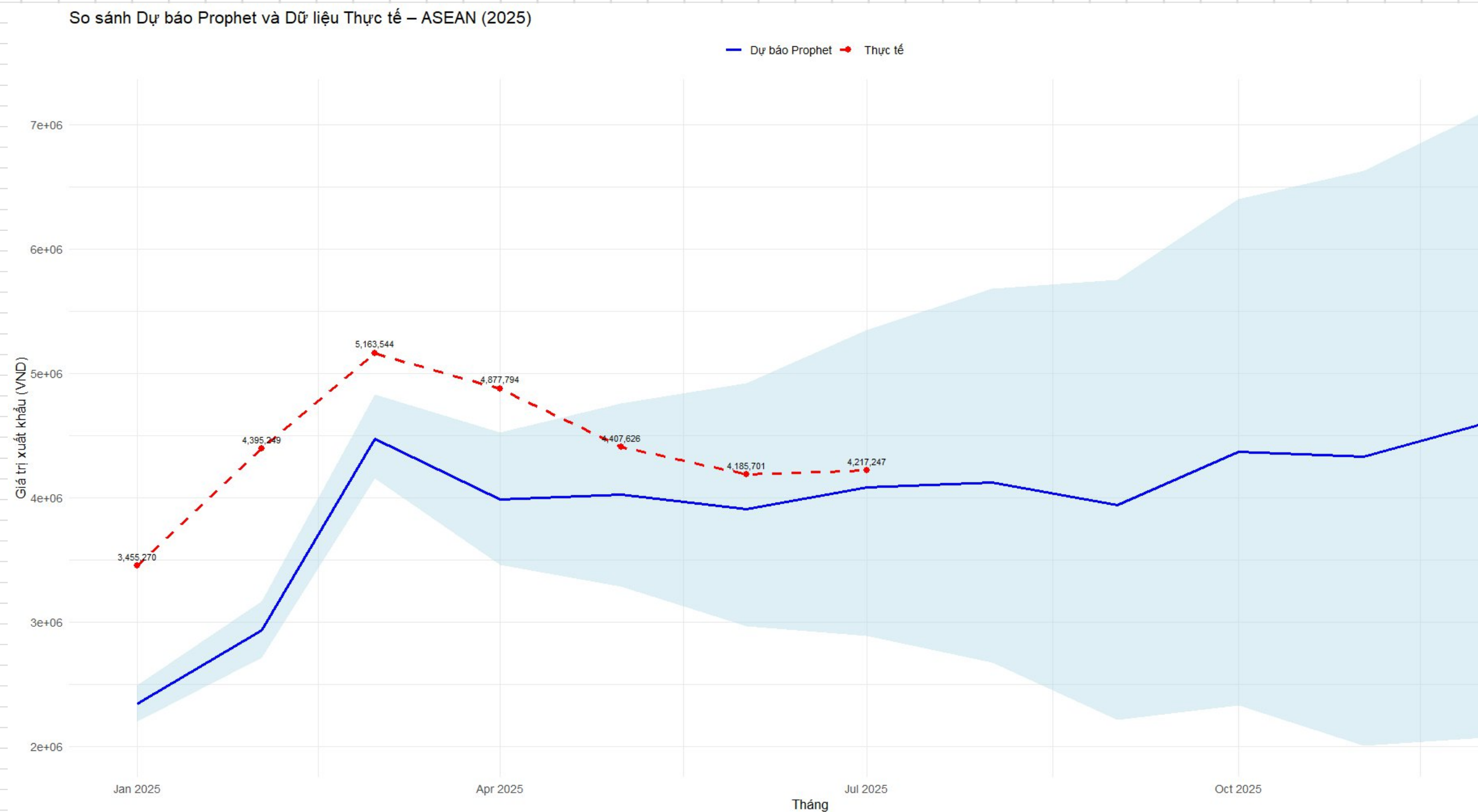
Implement Model

ASEAN



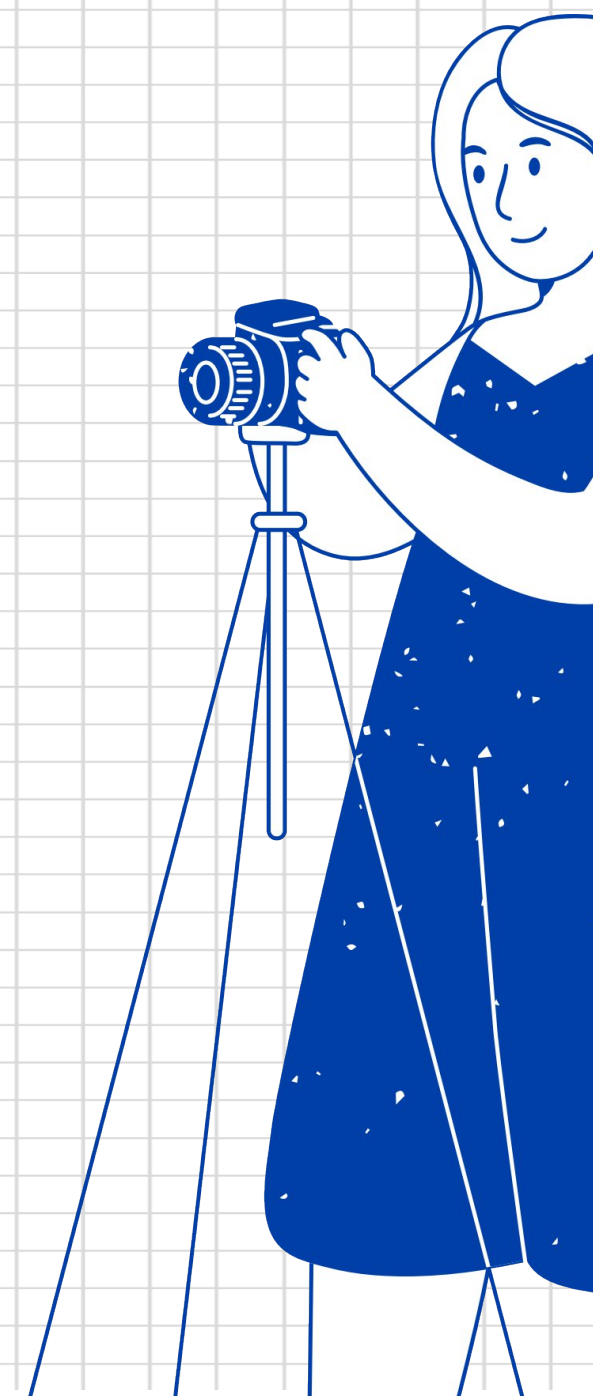
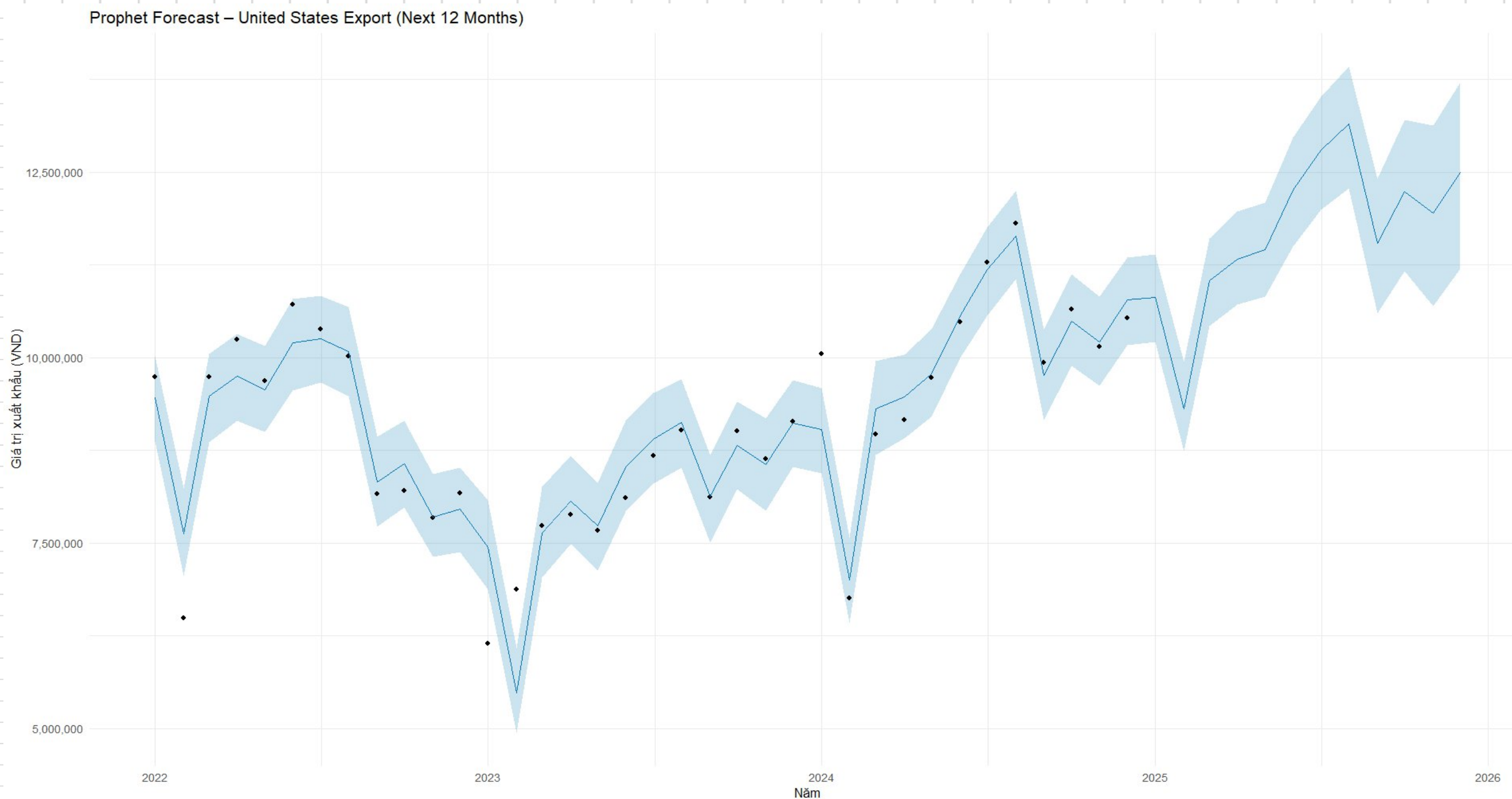
Implement Model

ASEAN



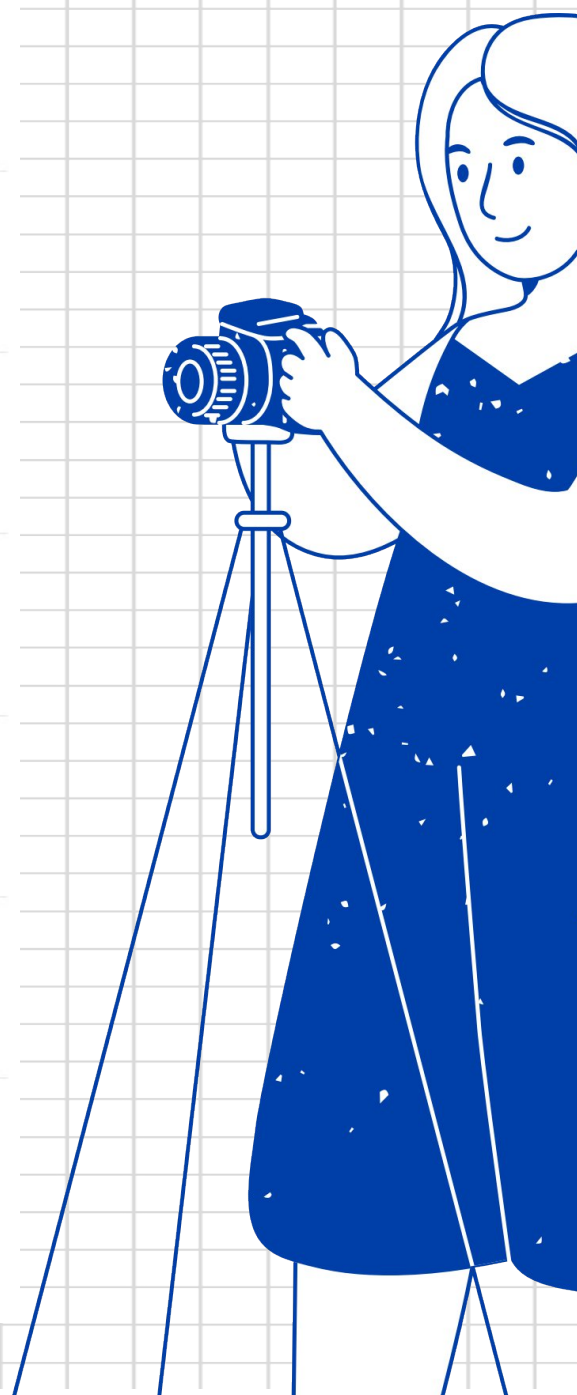
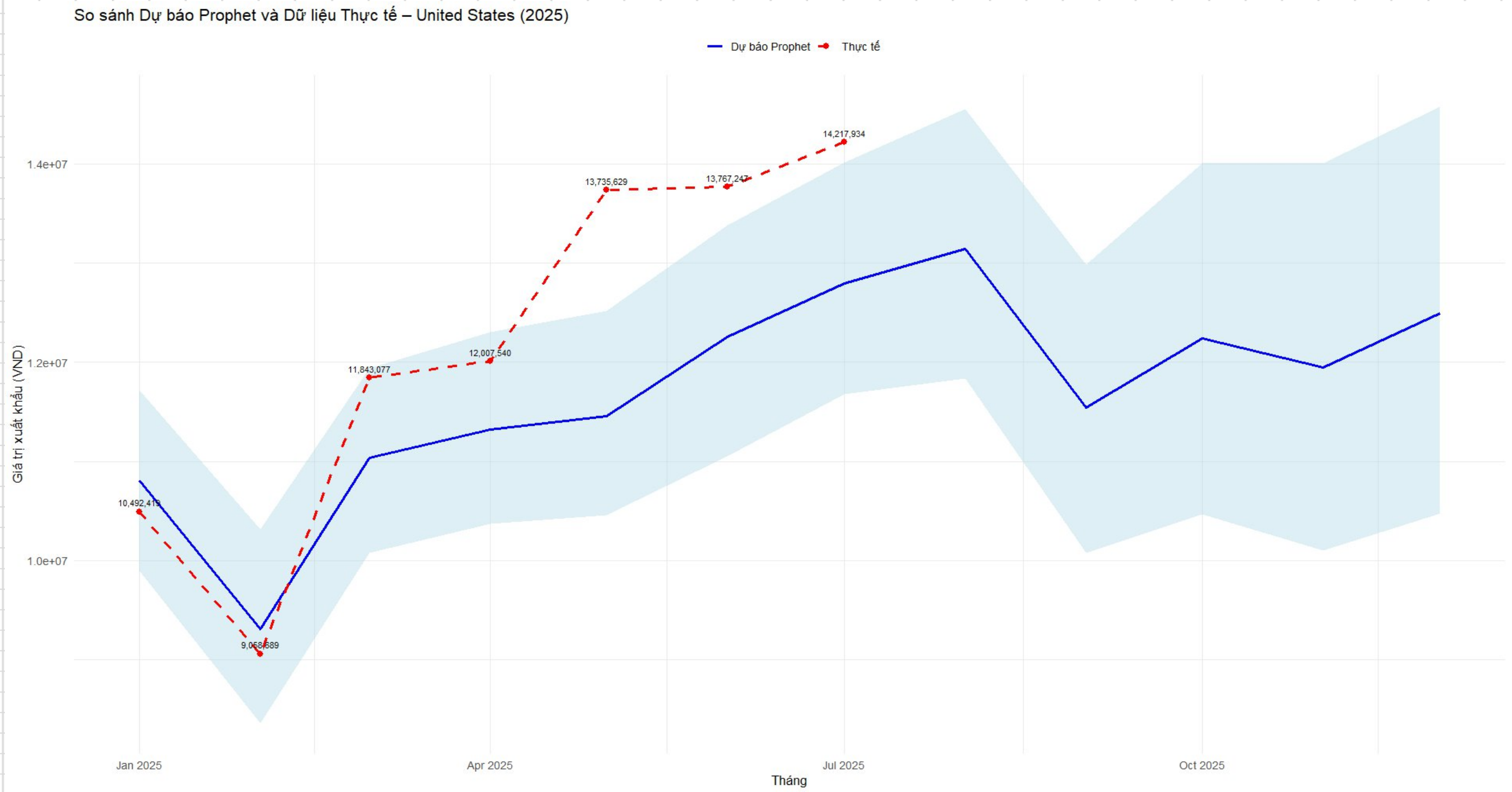
Implement Model

US



Implement Model

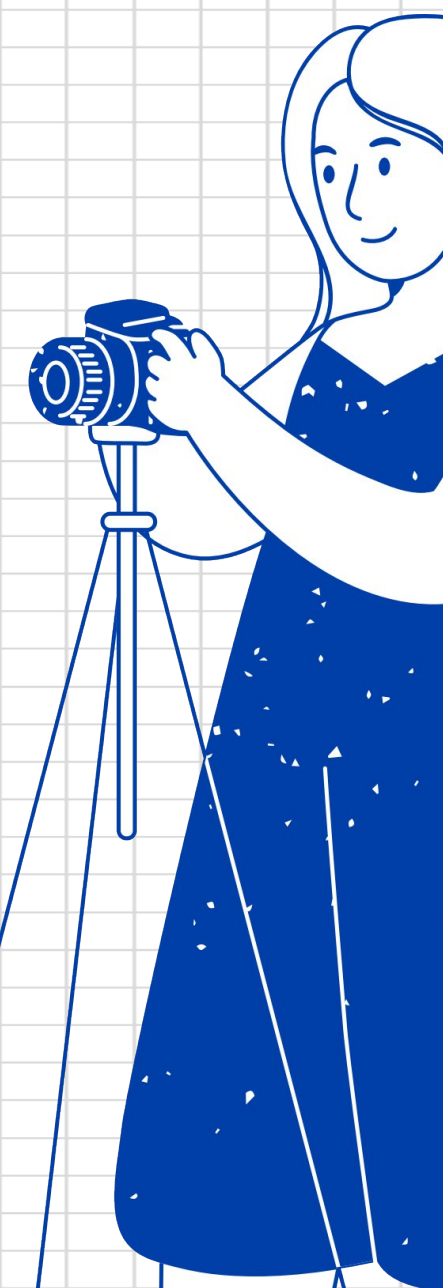
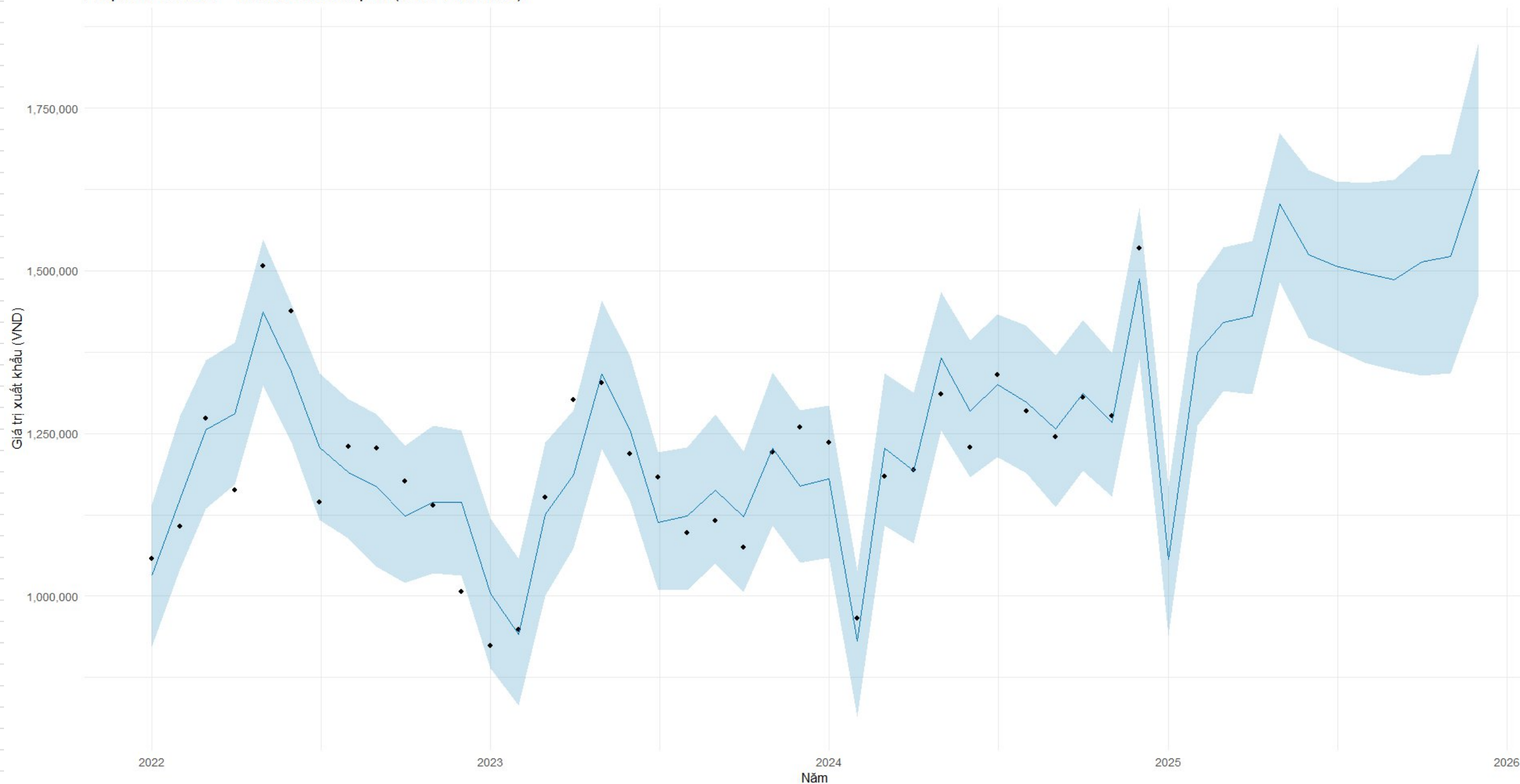
US



Implement Model

US

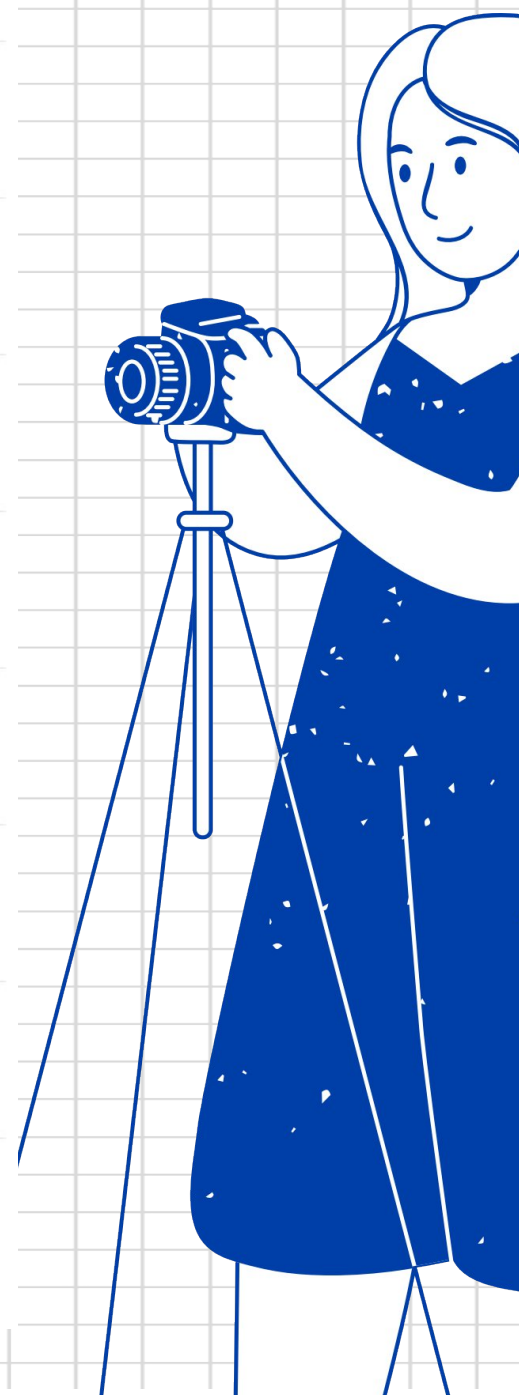
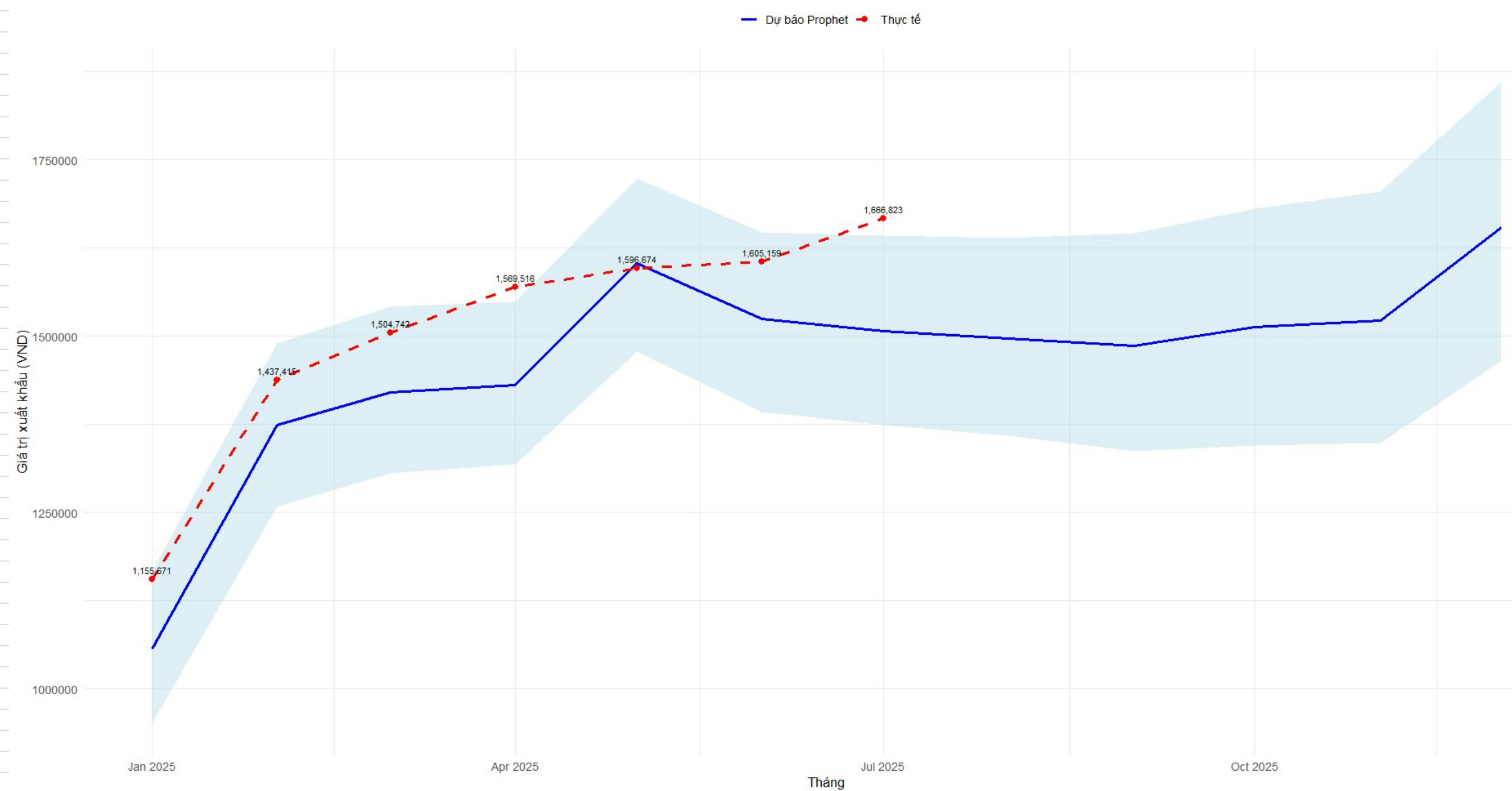
Prophet Forecast – United States Import (Next 12 Months)



Implement Model

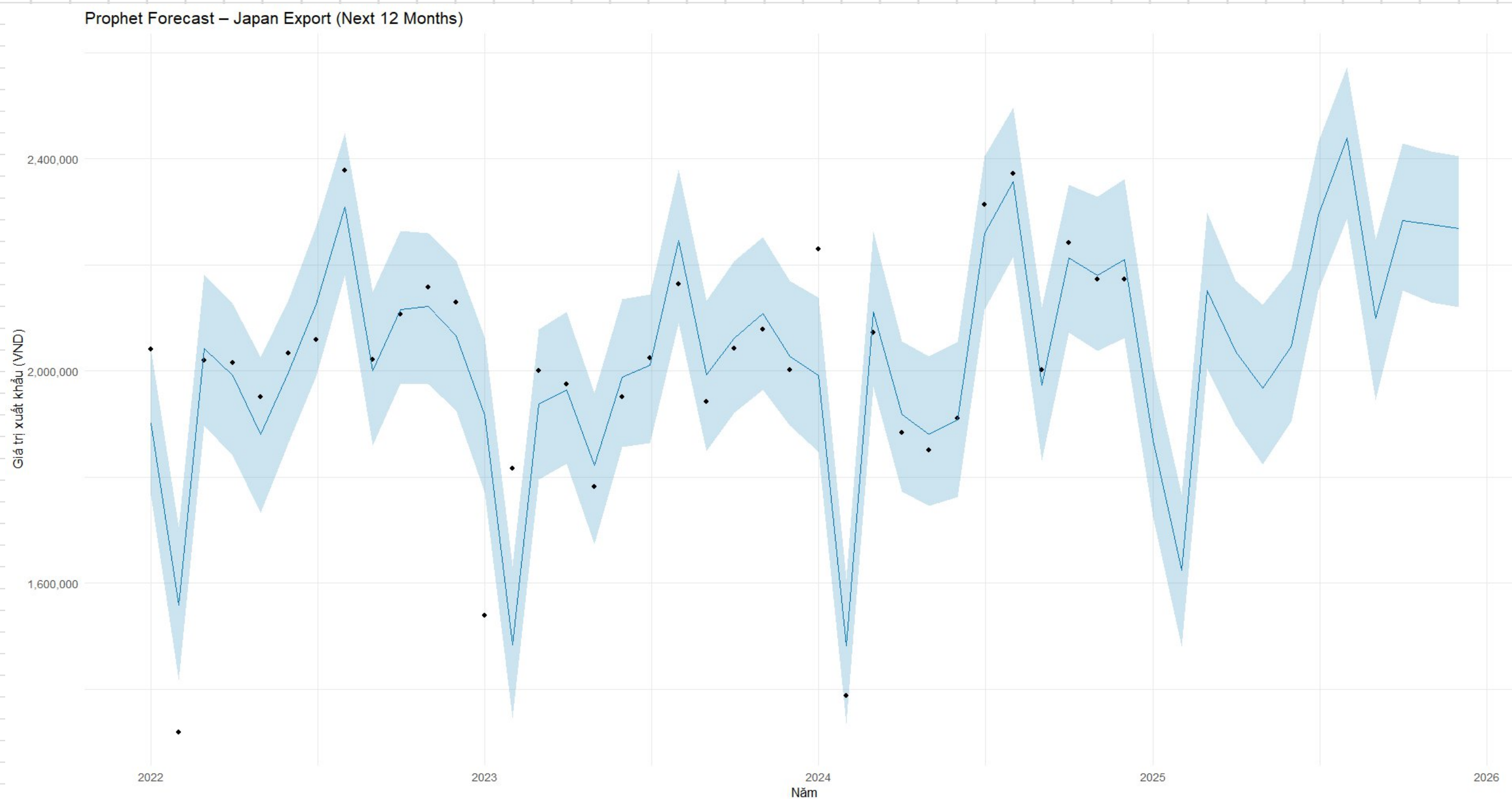
US

So sánh Dự báo Prophet và Dữ liệu Thực tế – United States (2025)



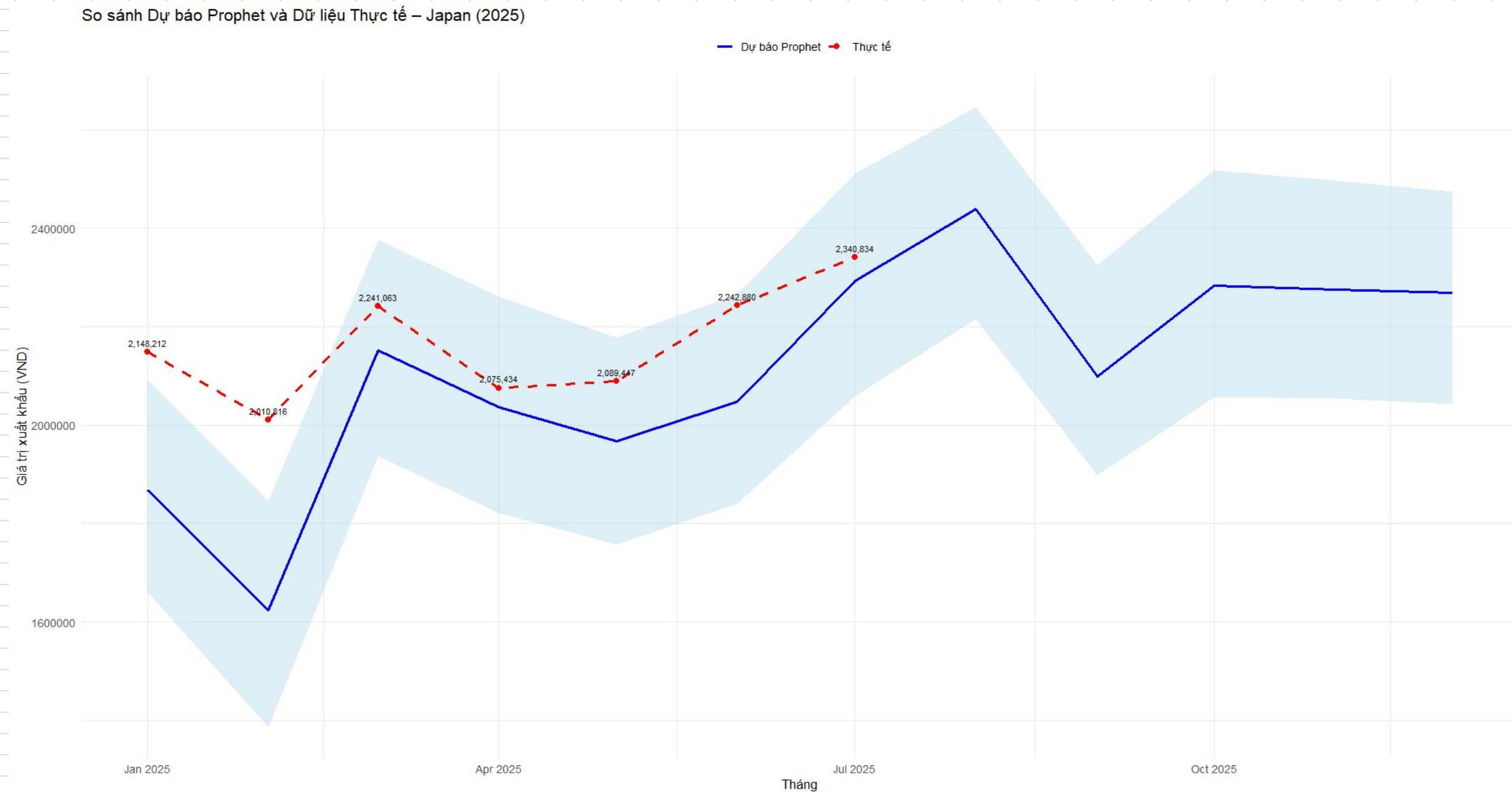
Implement Model

Japan



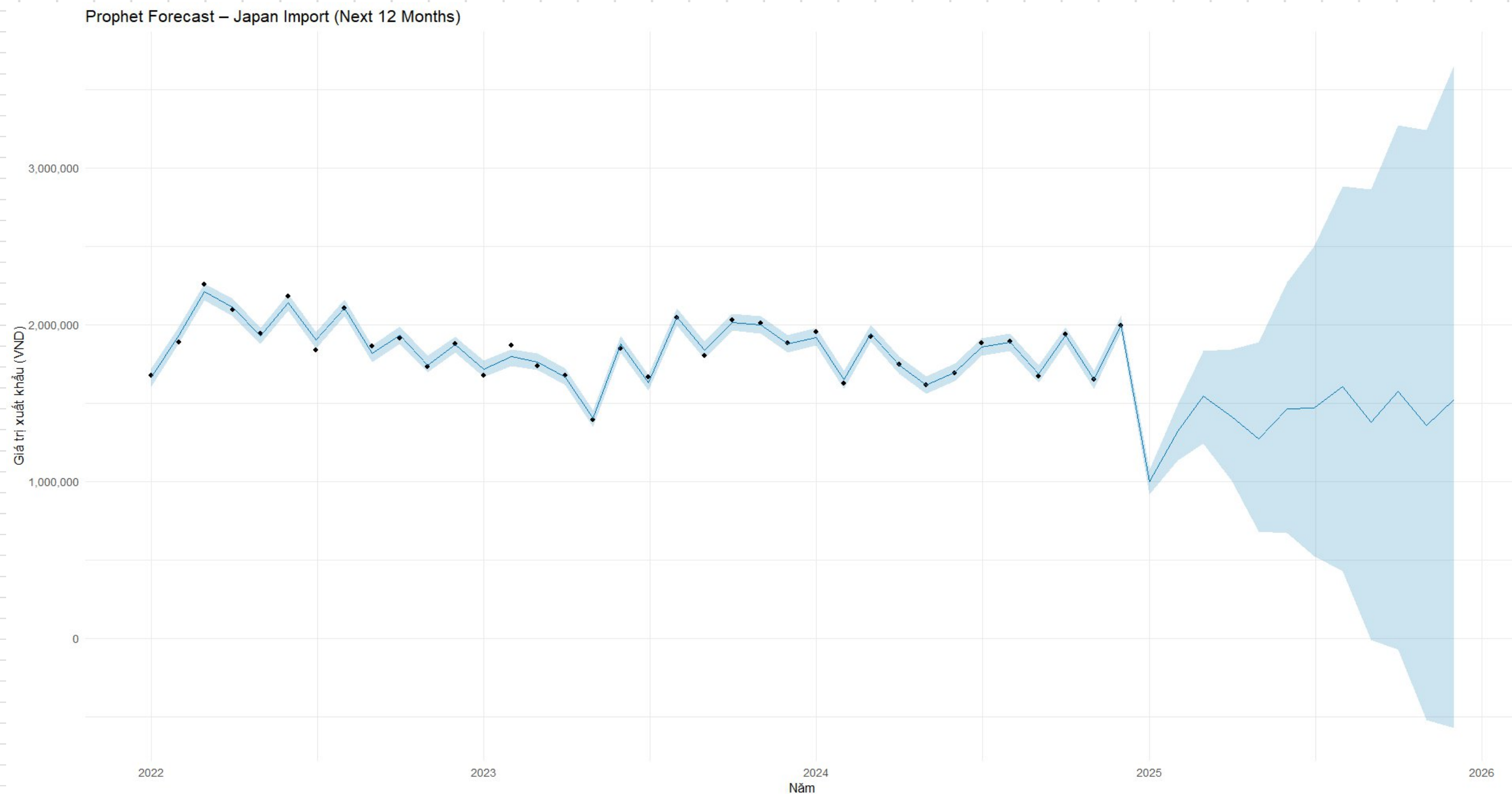
Implement Model

Japan



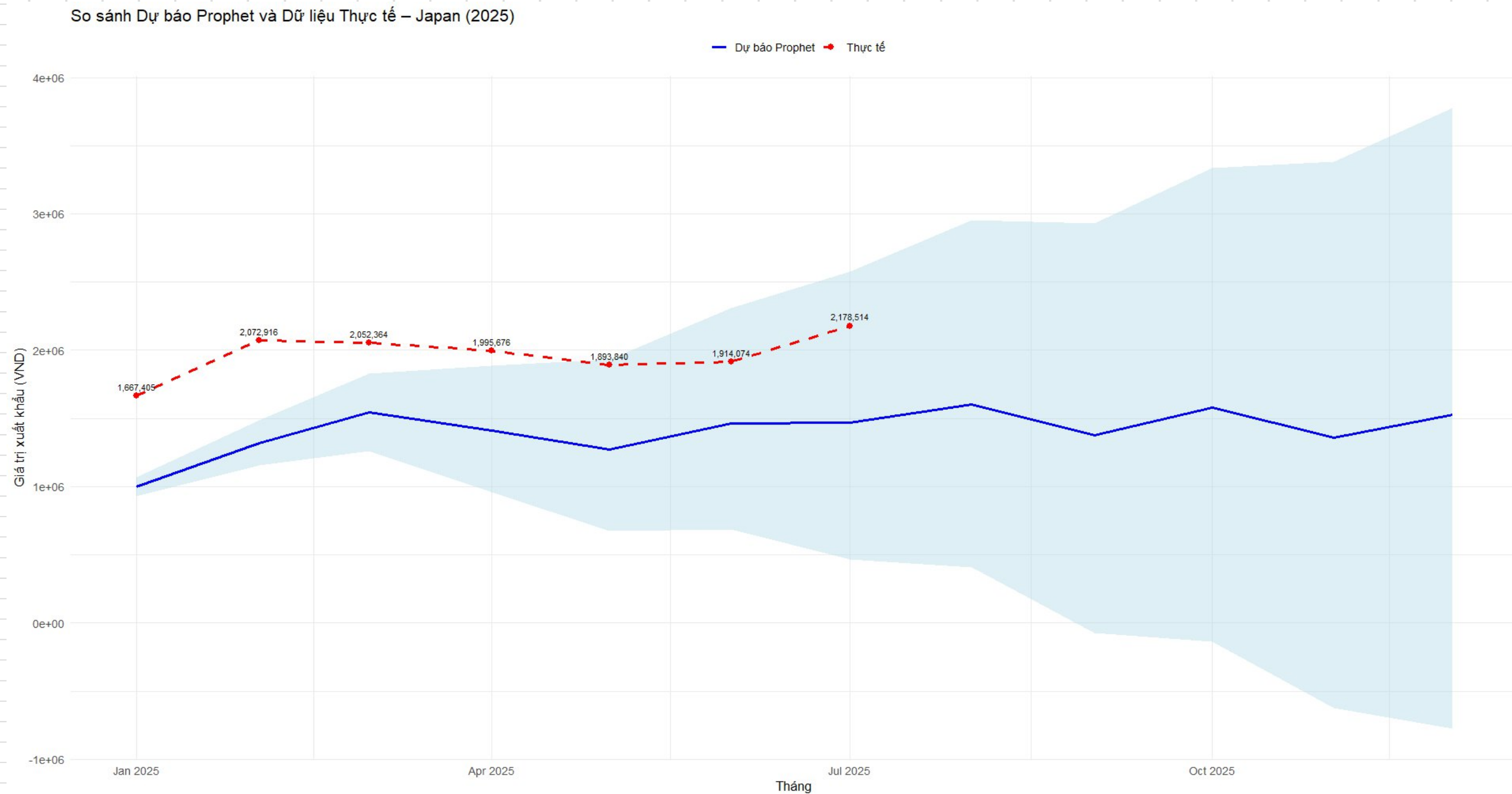
Implement Model

Japan



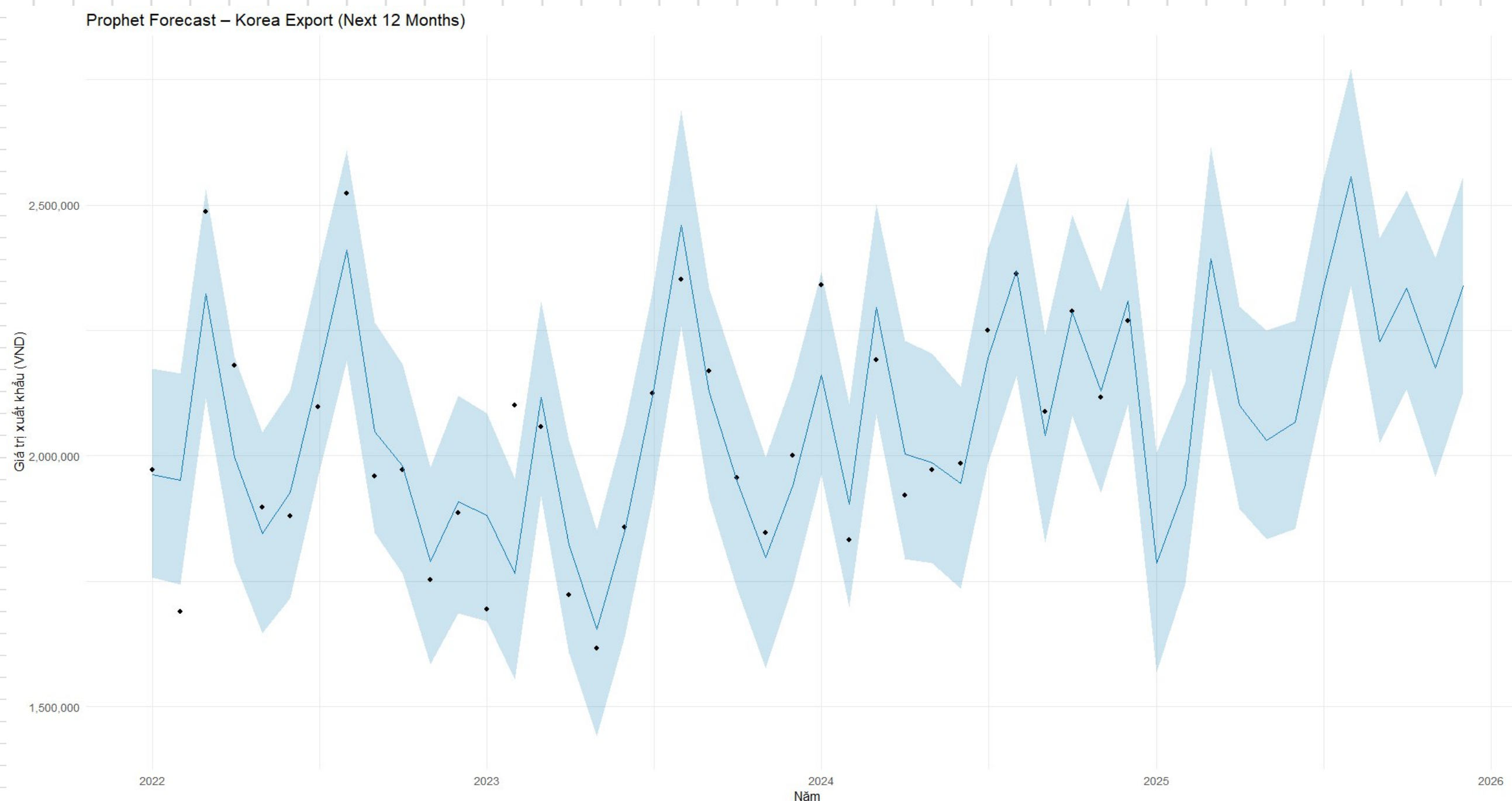
Implement Model

Japan



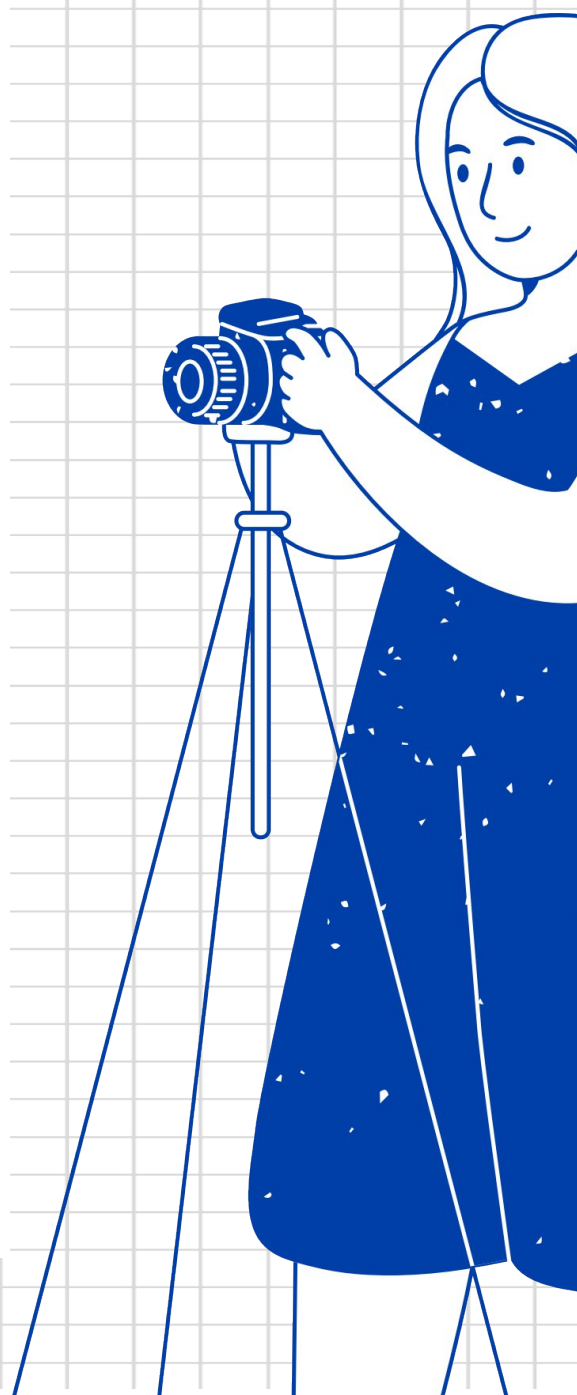
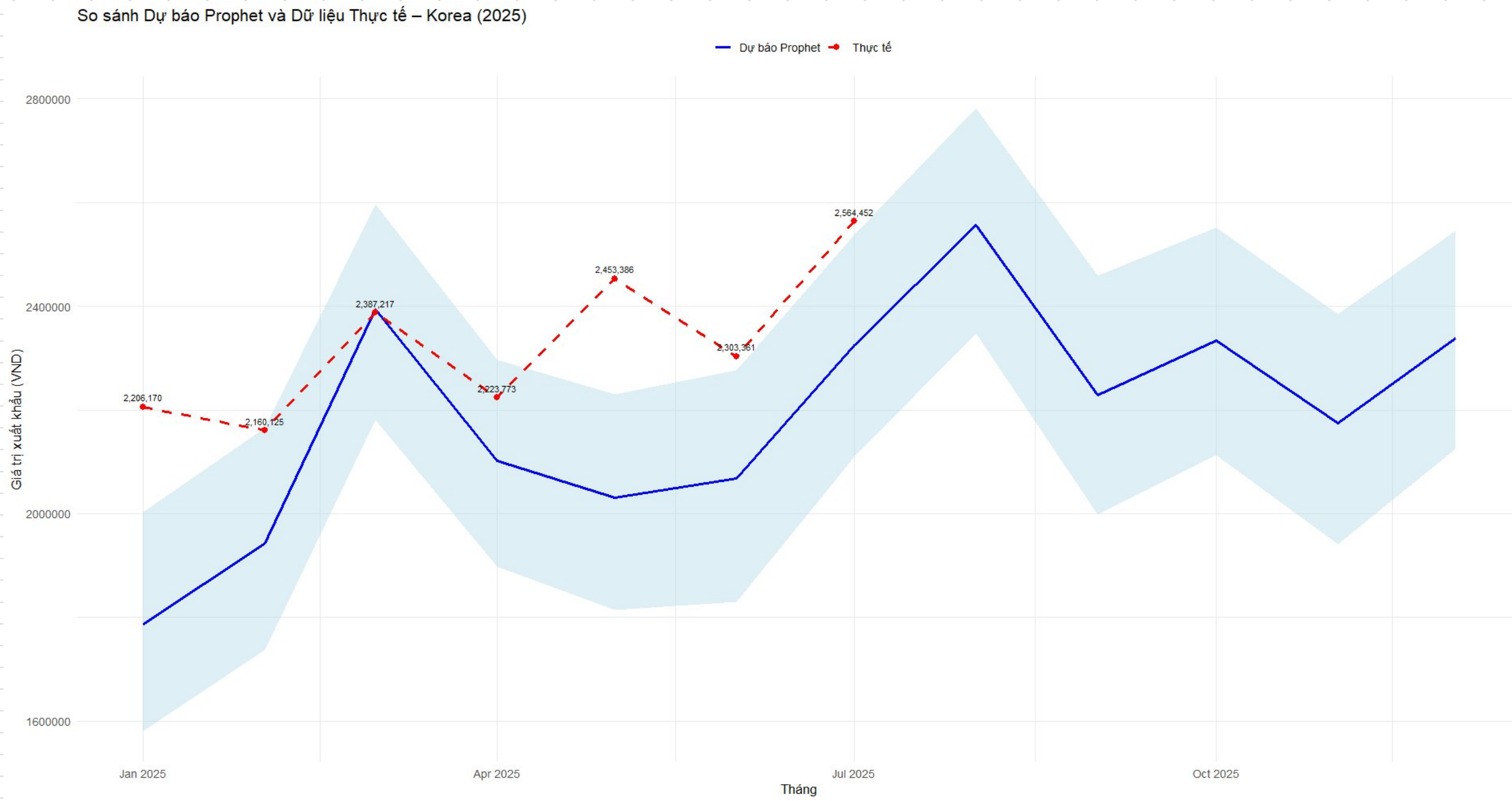
Implement Model

Korea



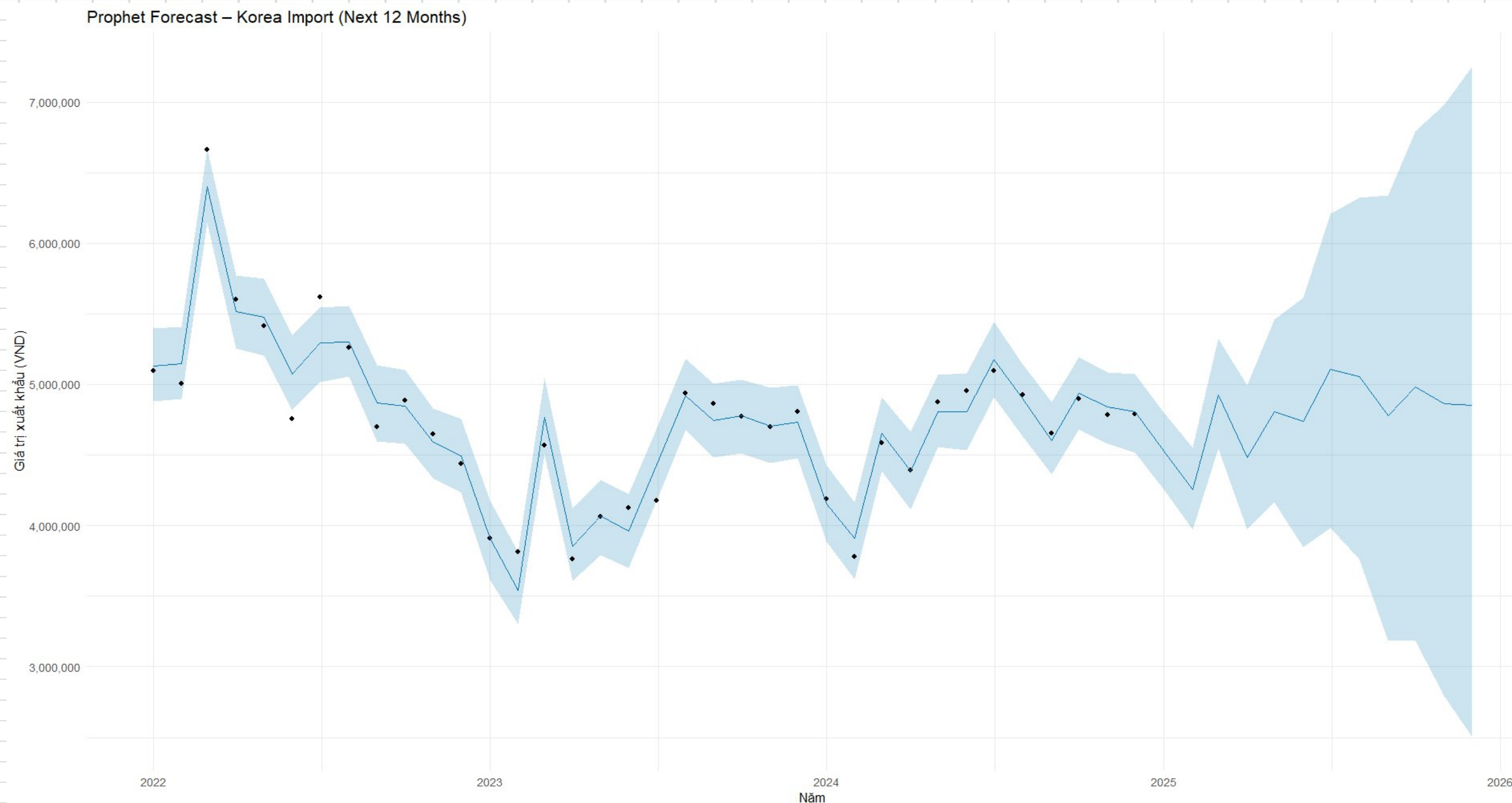
Implement Model

Korea



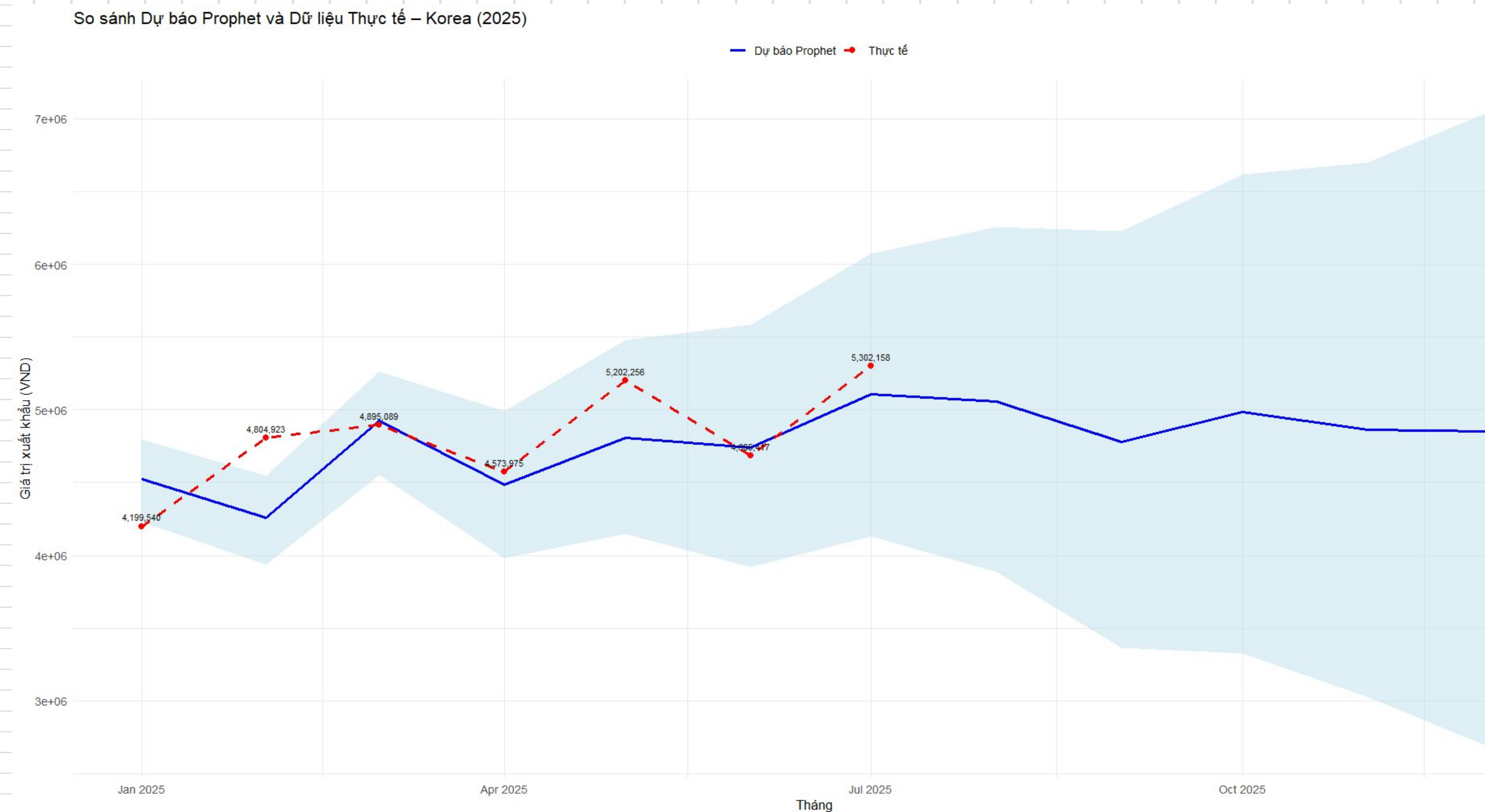
Implement Model

Korea



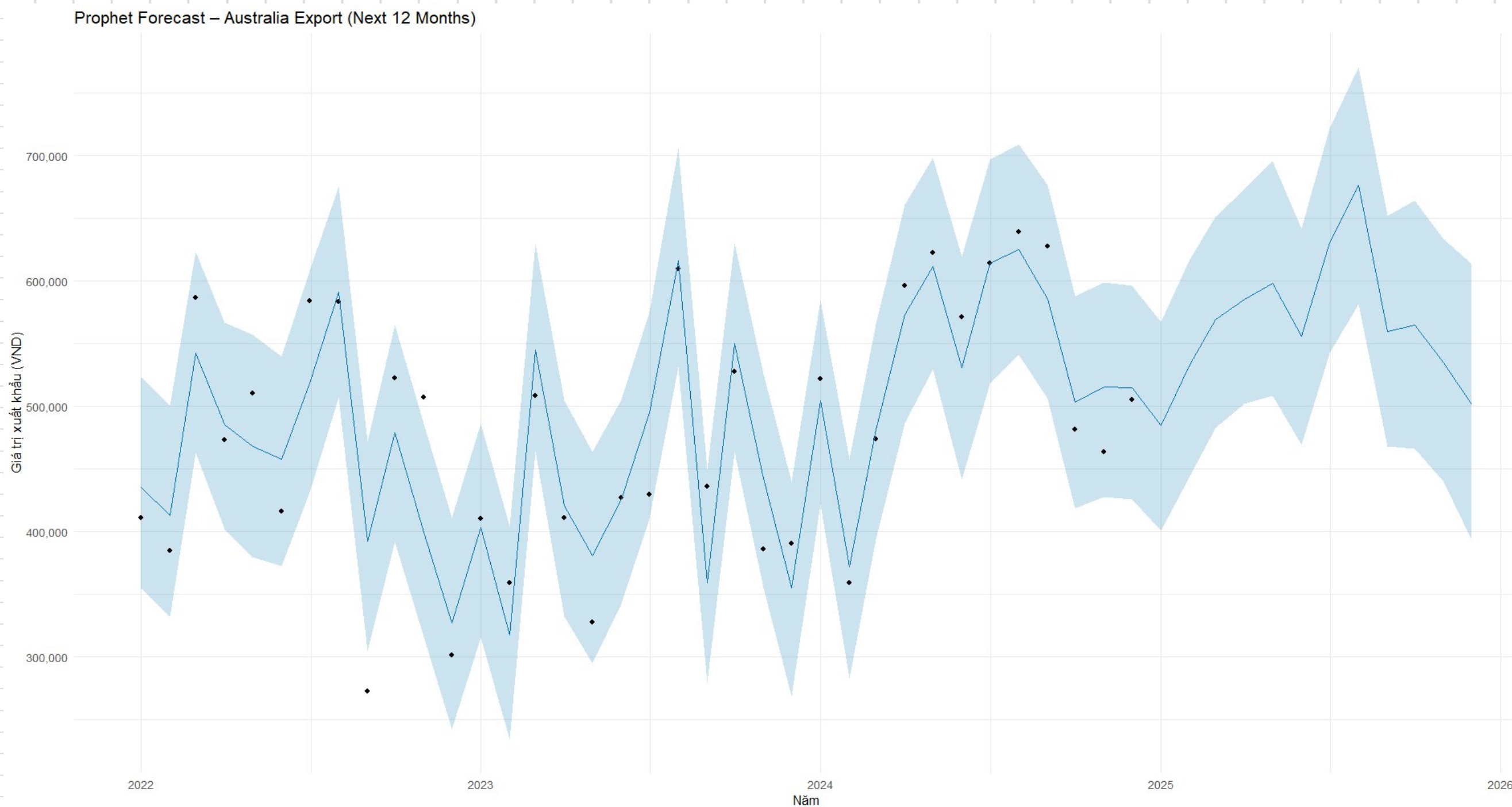
Implement Model

Korea



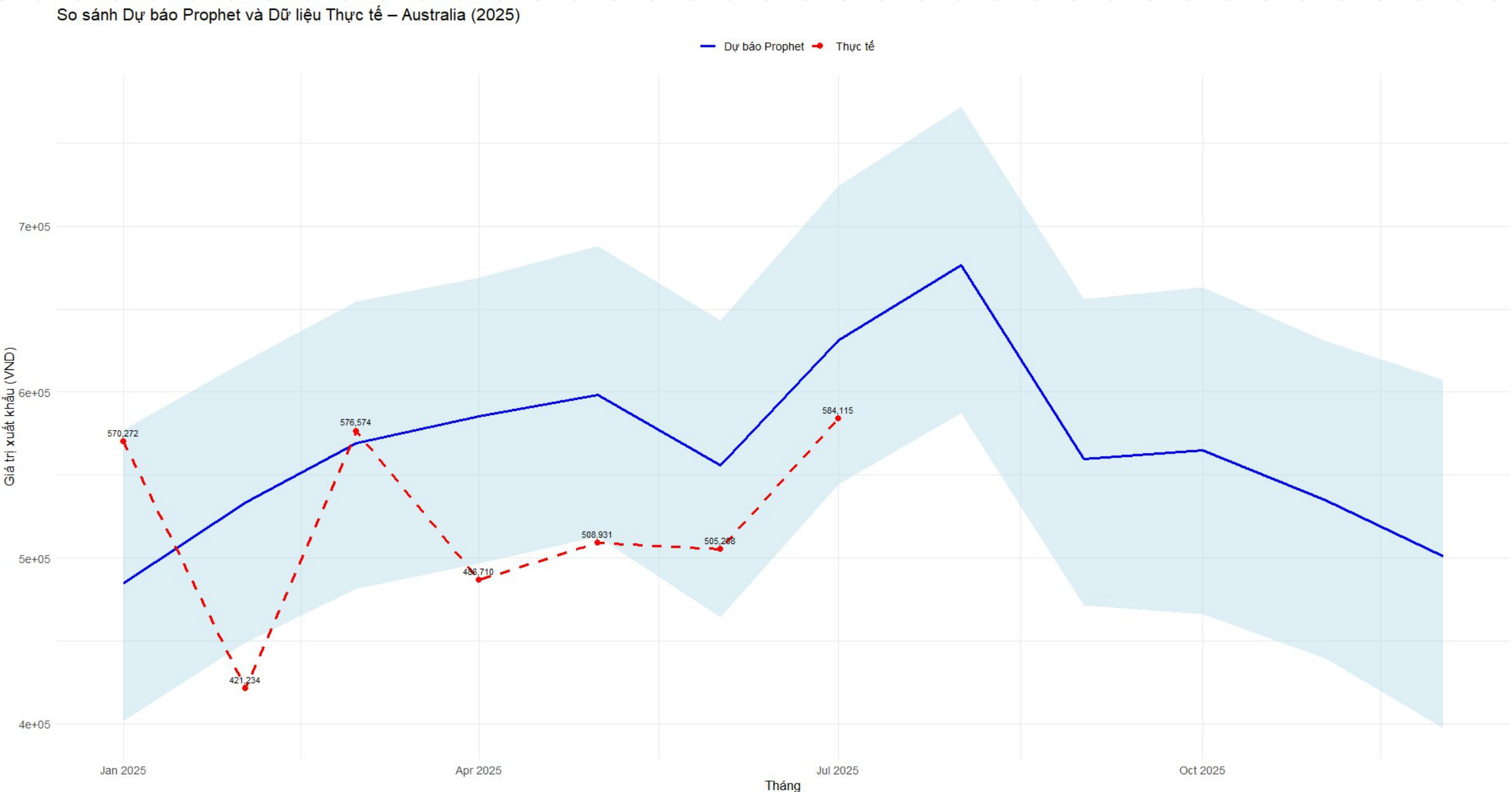
Implement Model

Australia



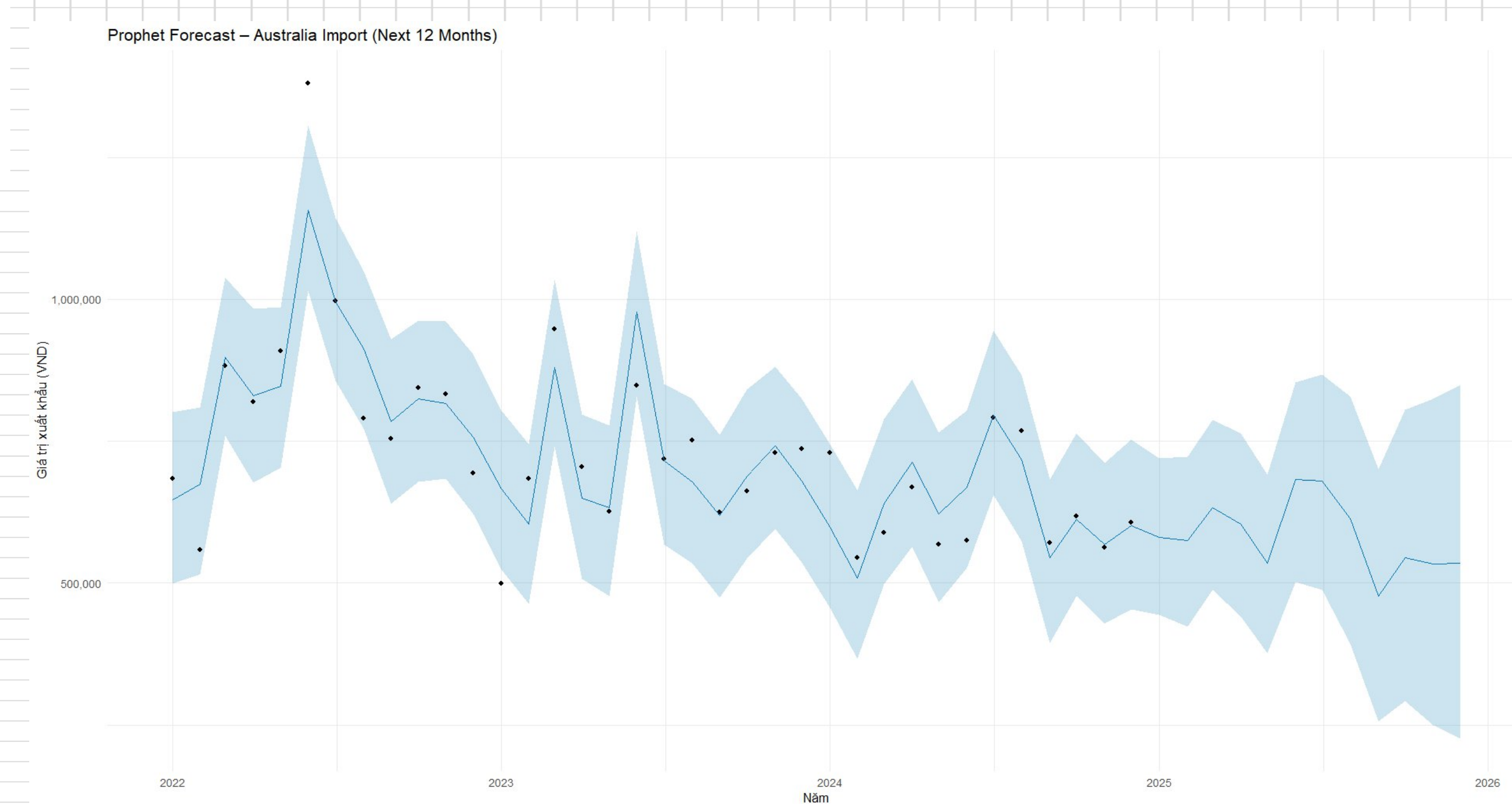
Implement Model

Australia



Implement Model

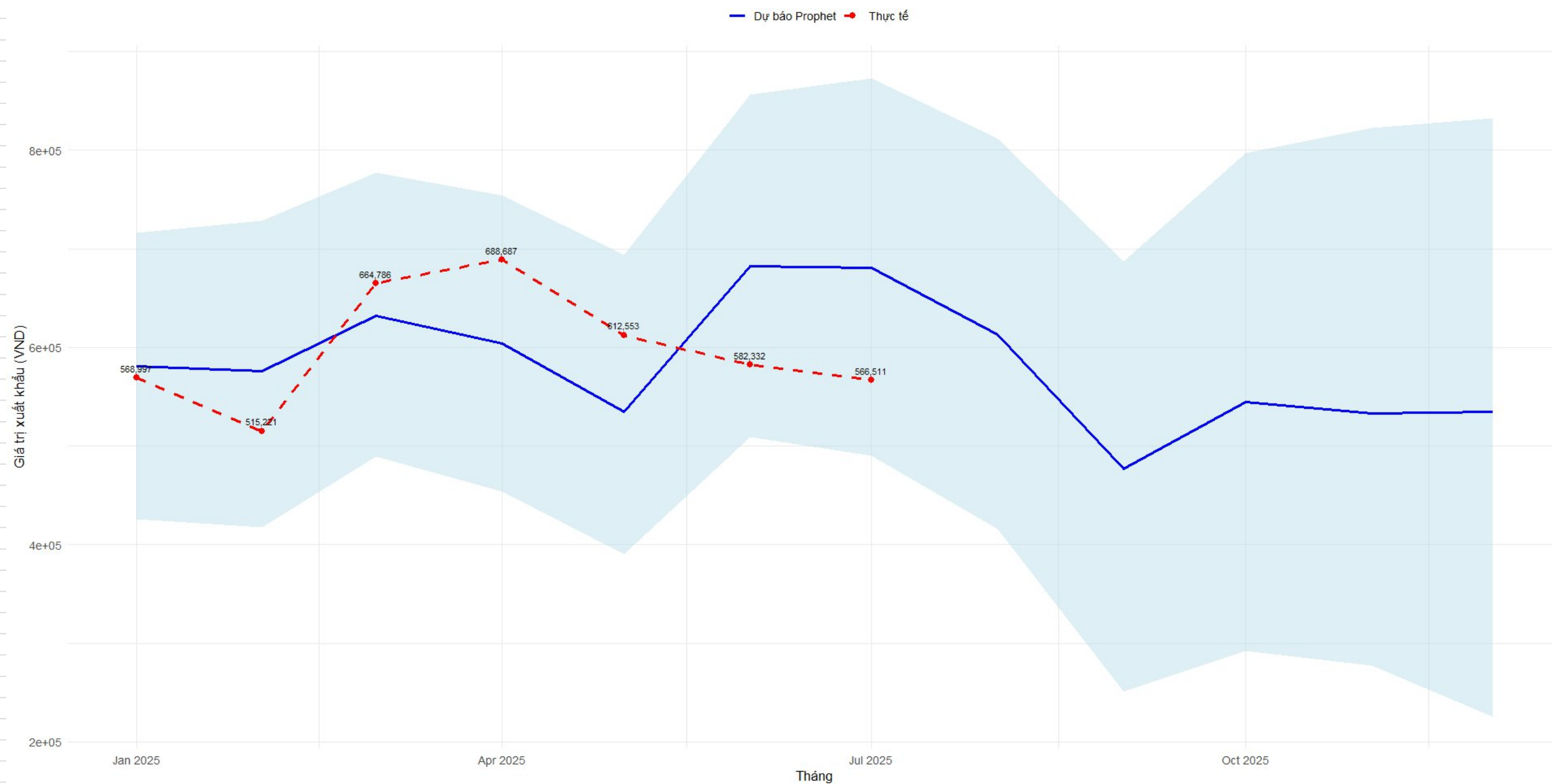
Australia



Implement Model

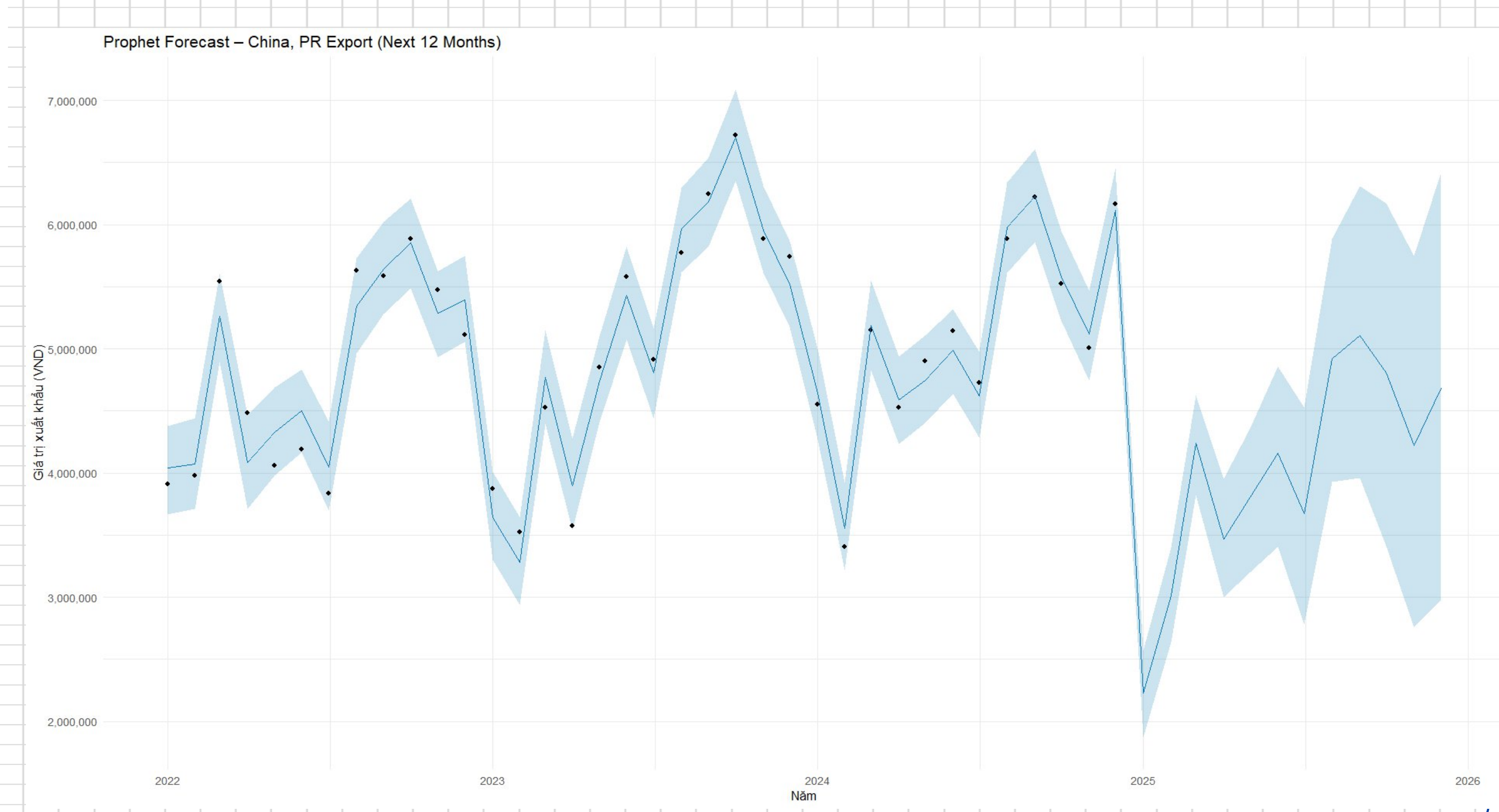
Australia

So sánh Dự báo Prophet và Dữ liệu Thực tế – Australia (2025)



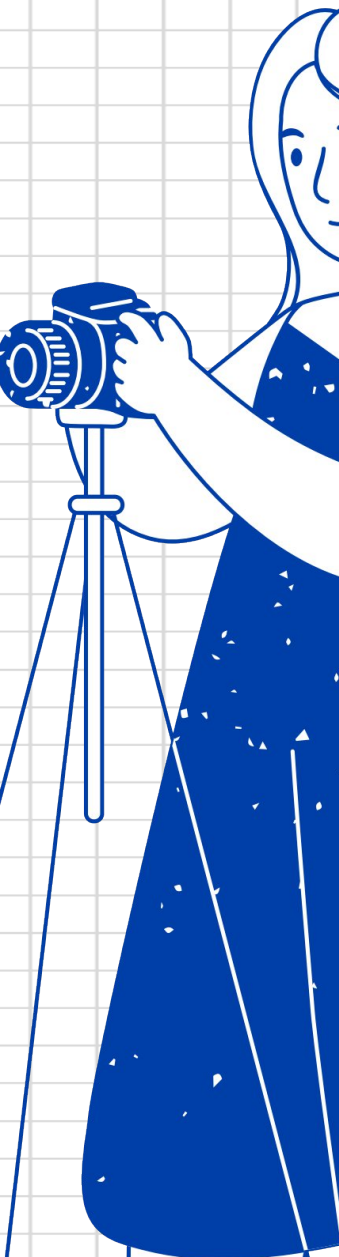
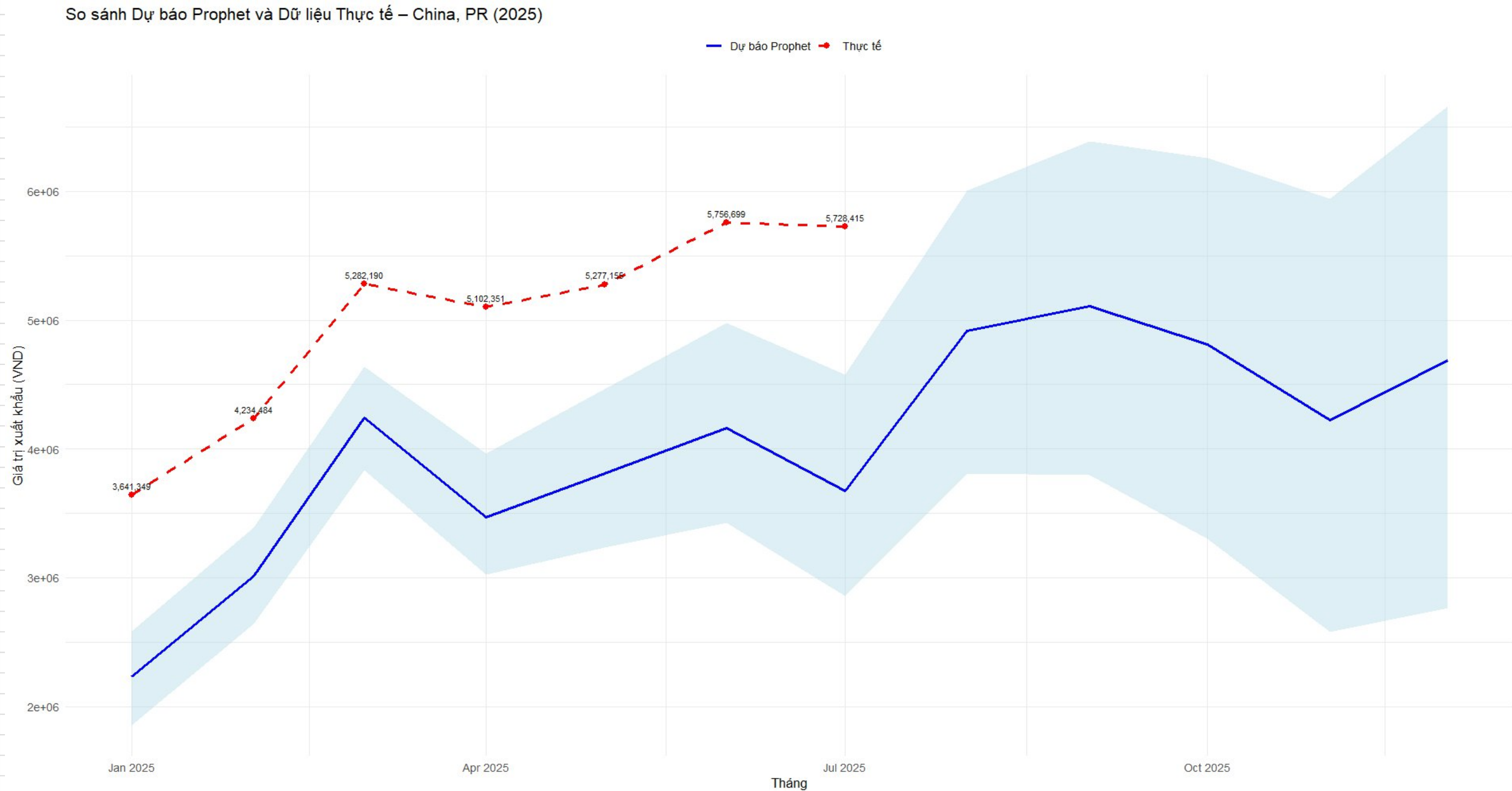
Implement Model

China



Implement Model

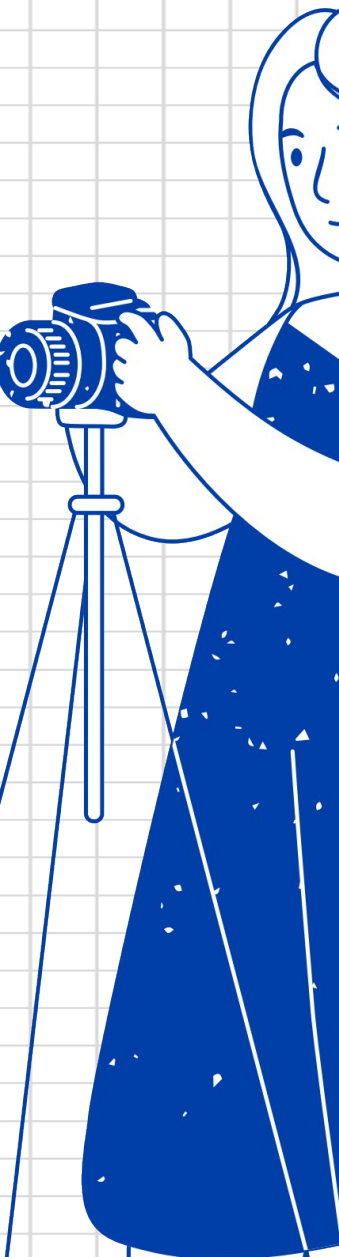
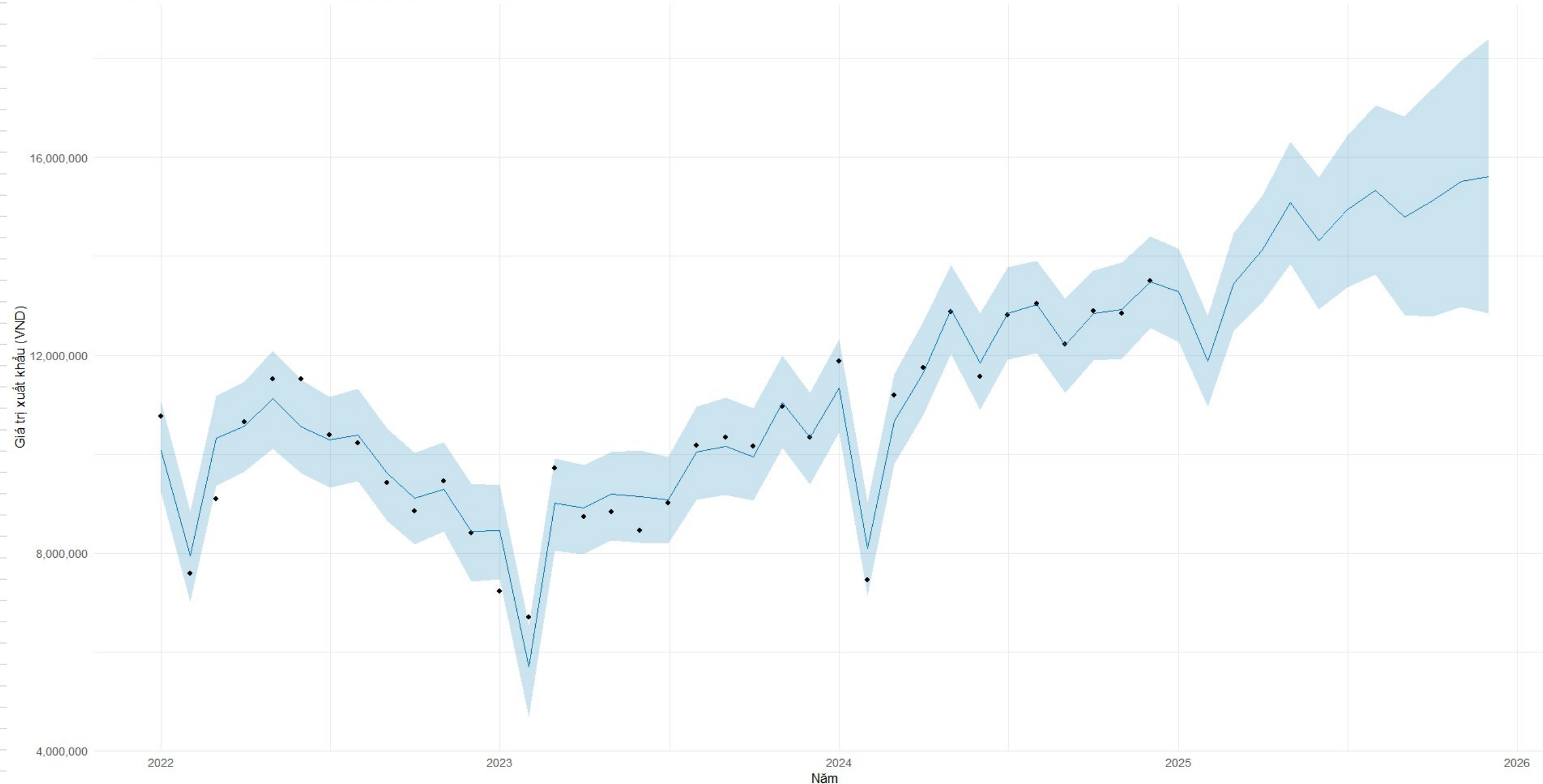
China



Implement Model

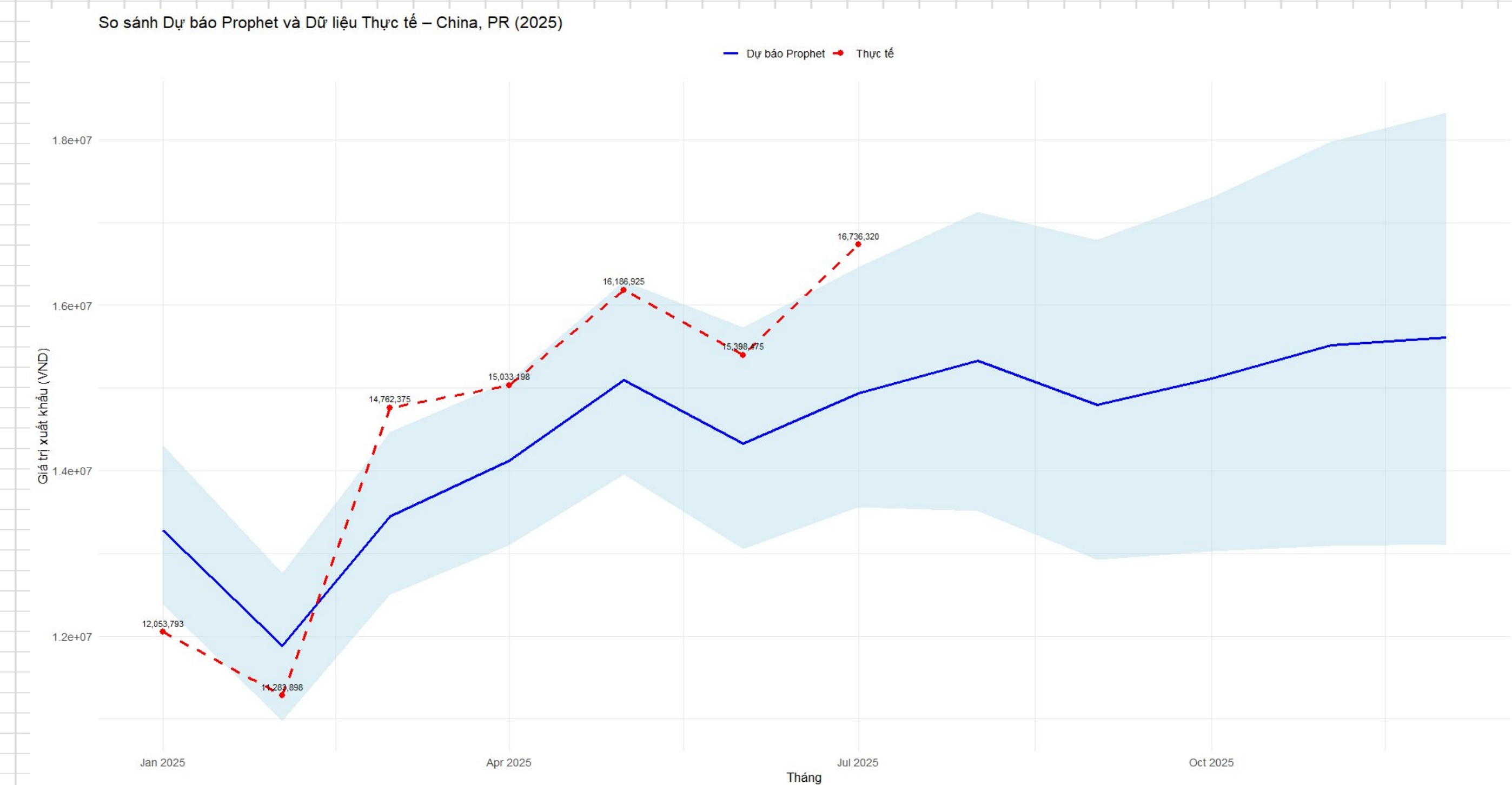
China

Prophet Forecast – China, PR Import (Next 12 Months)



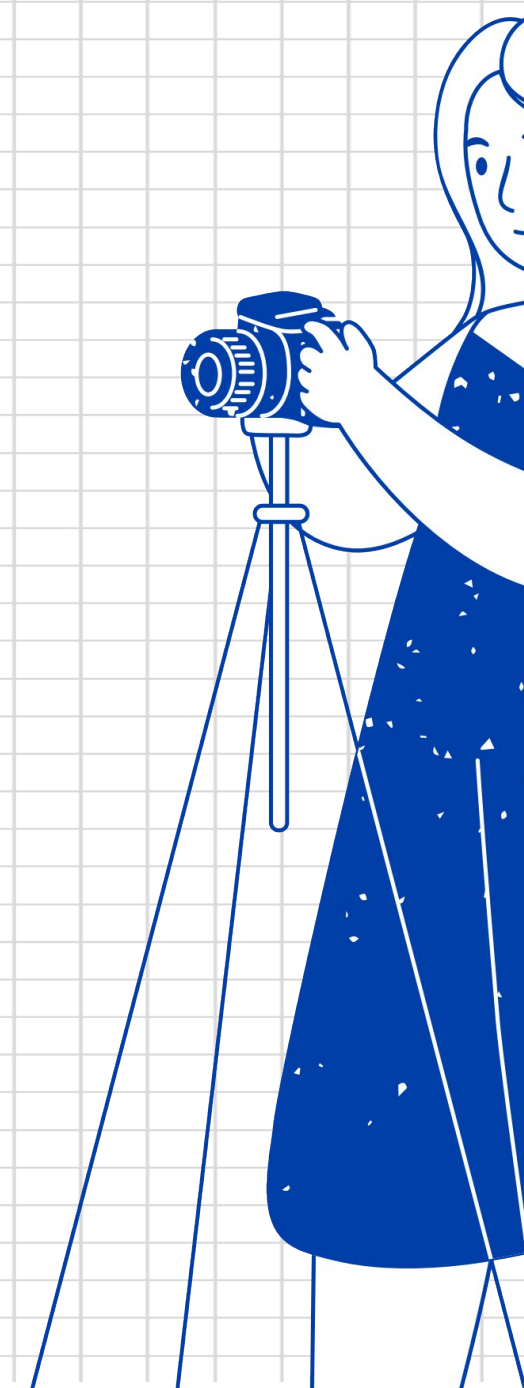
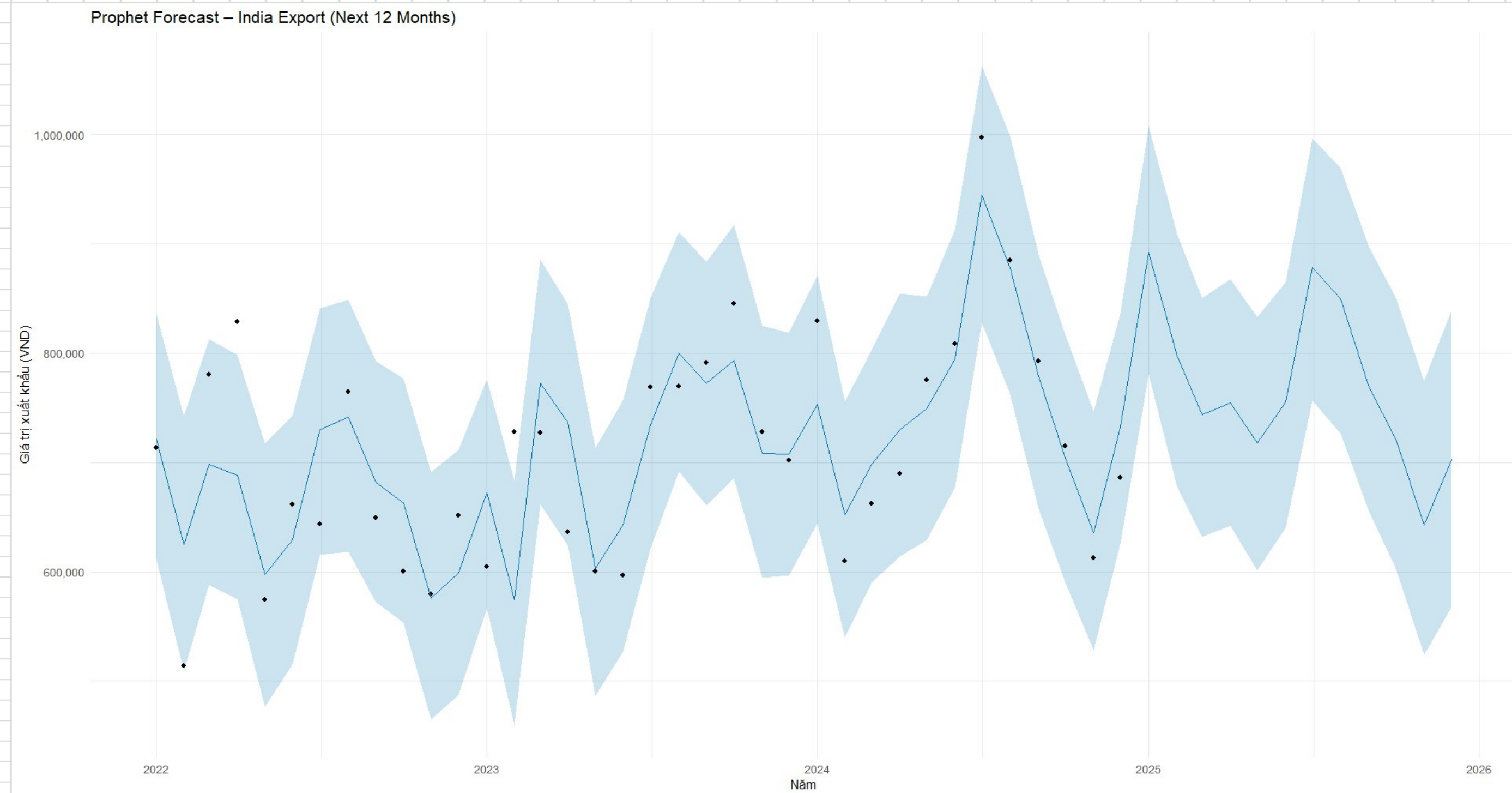
Implement Model

China



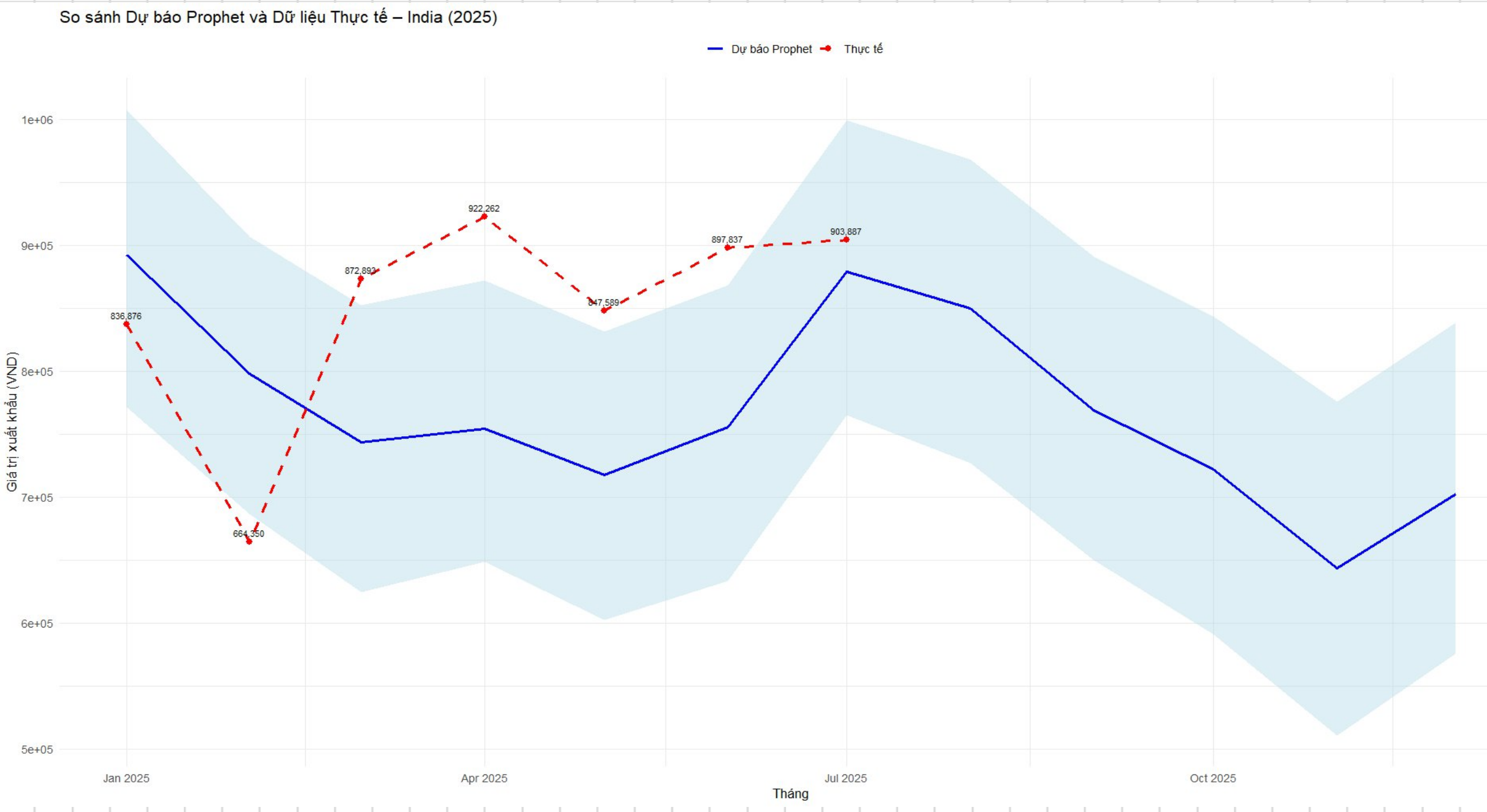
Implement Model

India



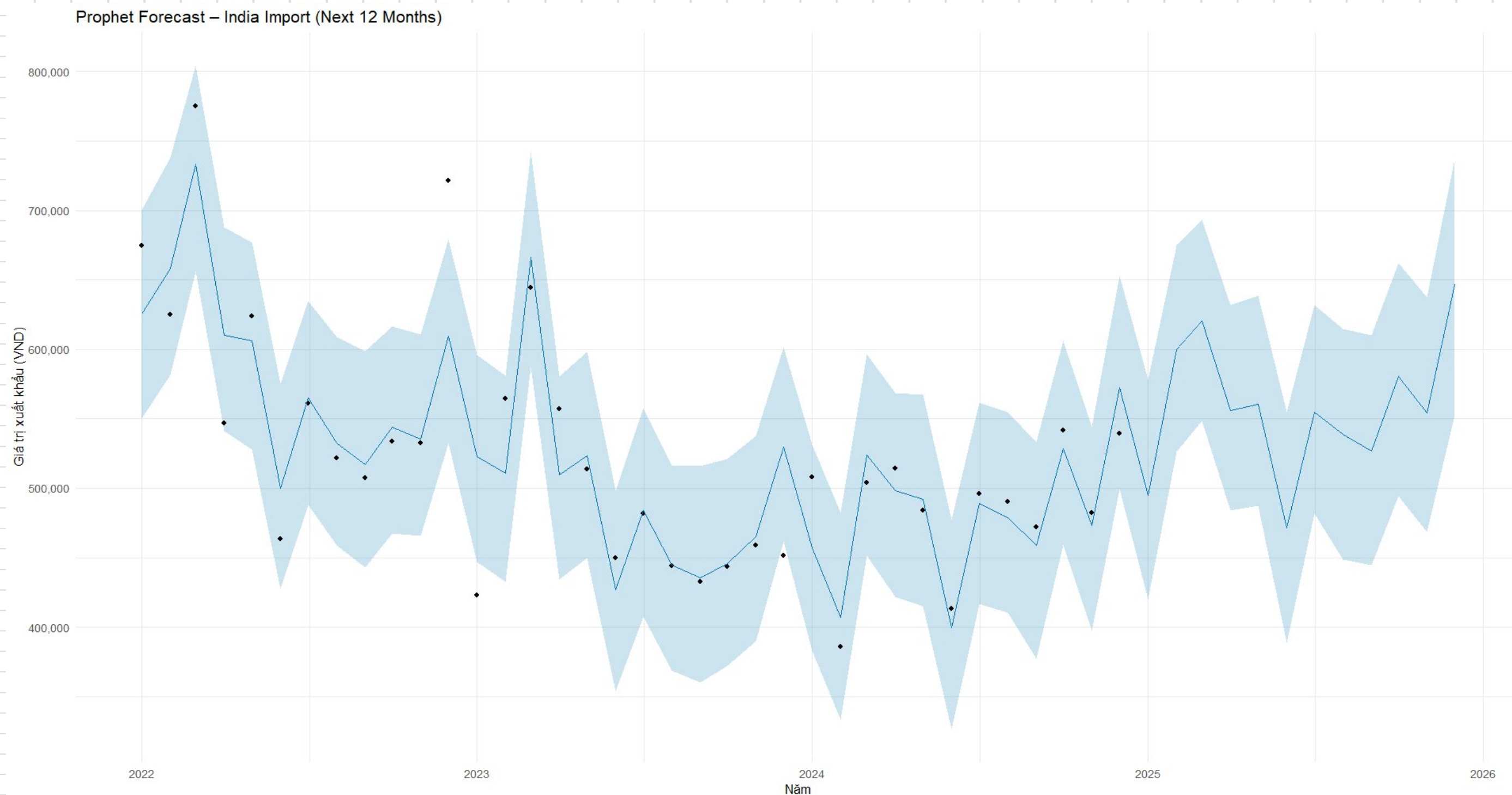
Implement Model

India



Implement Model

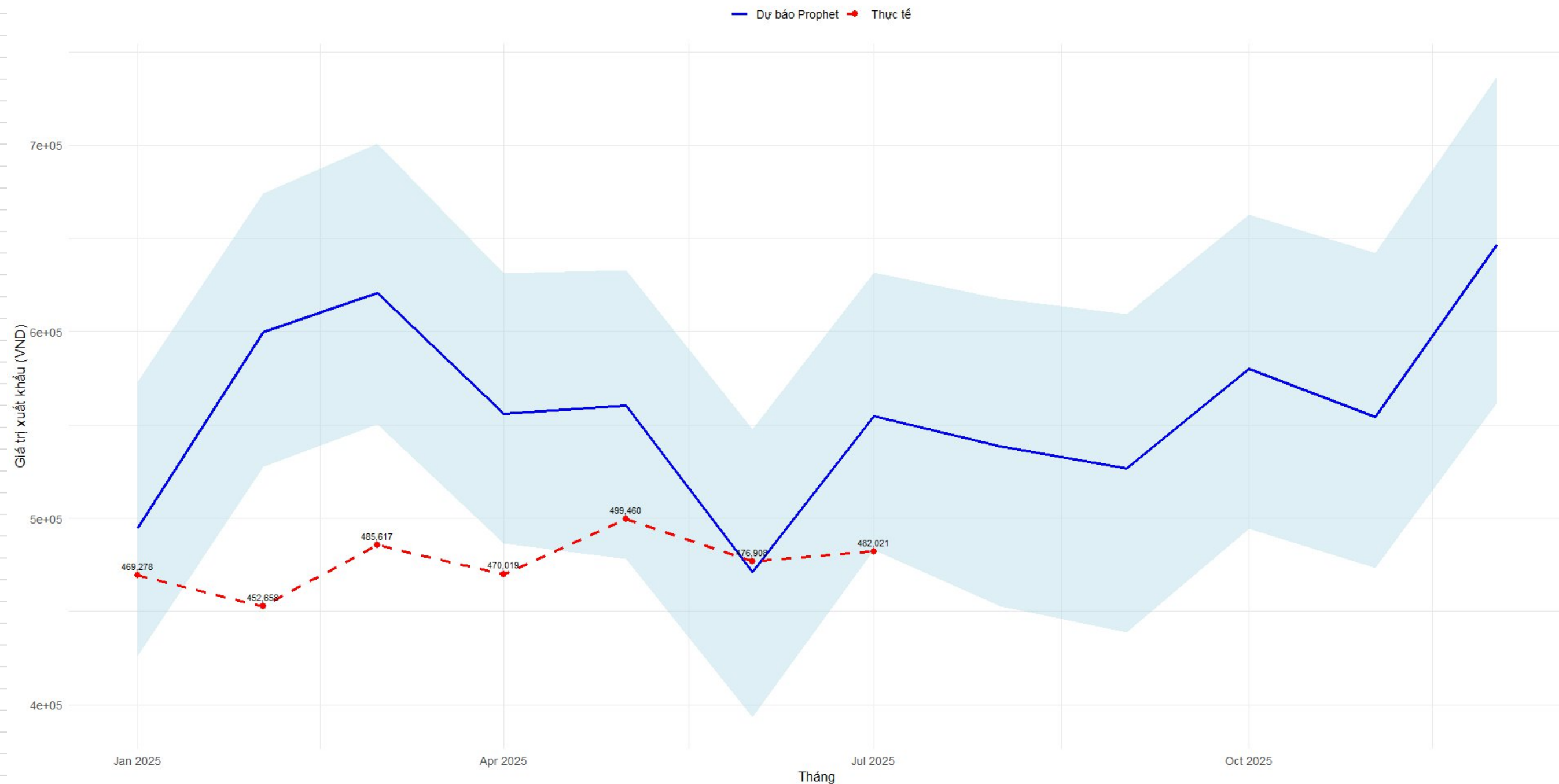
India



Implement Model

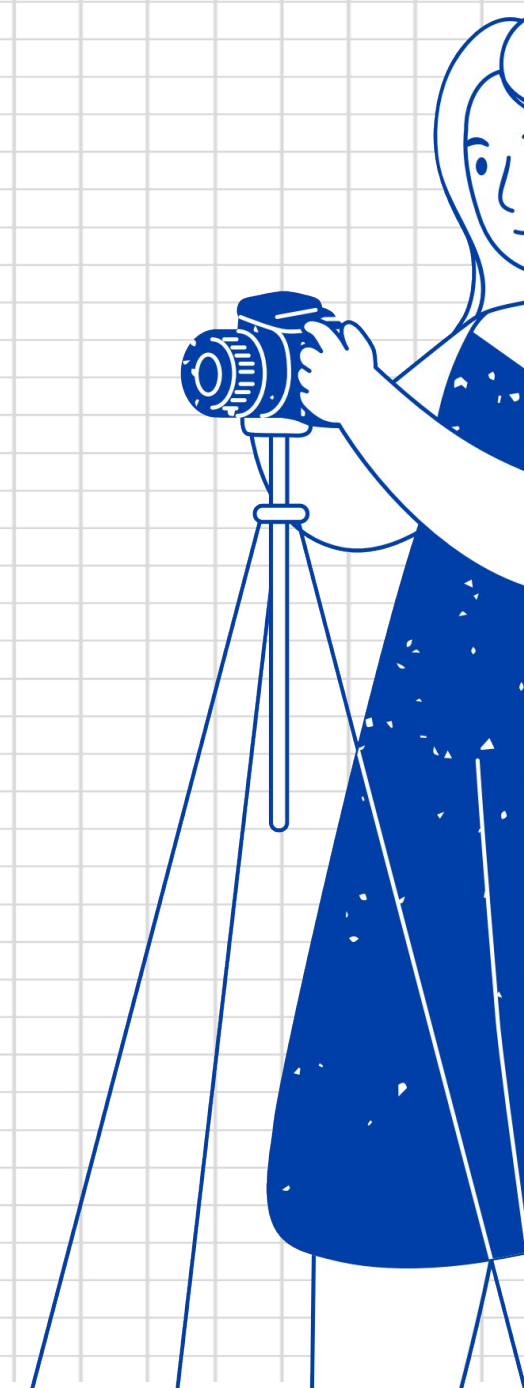
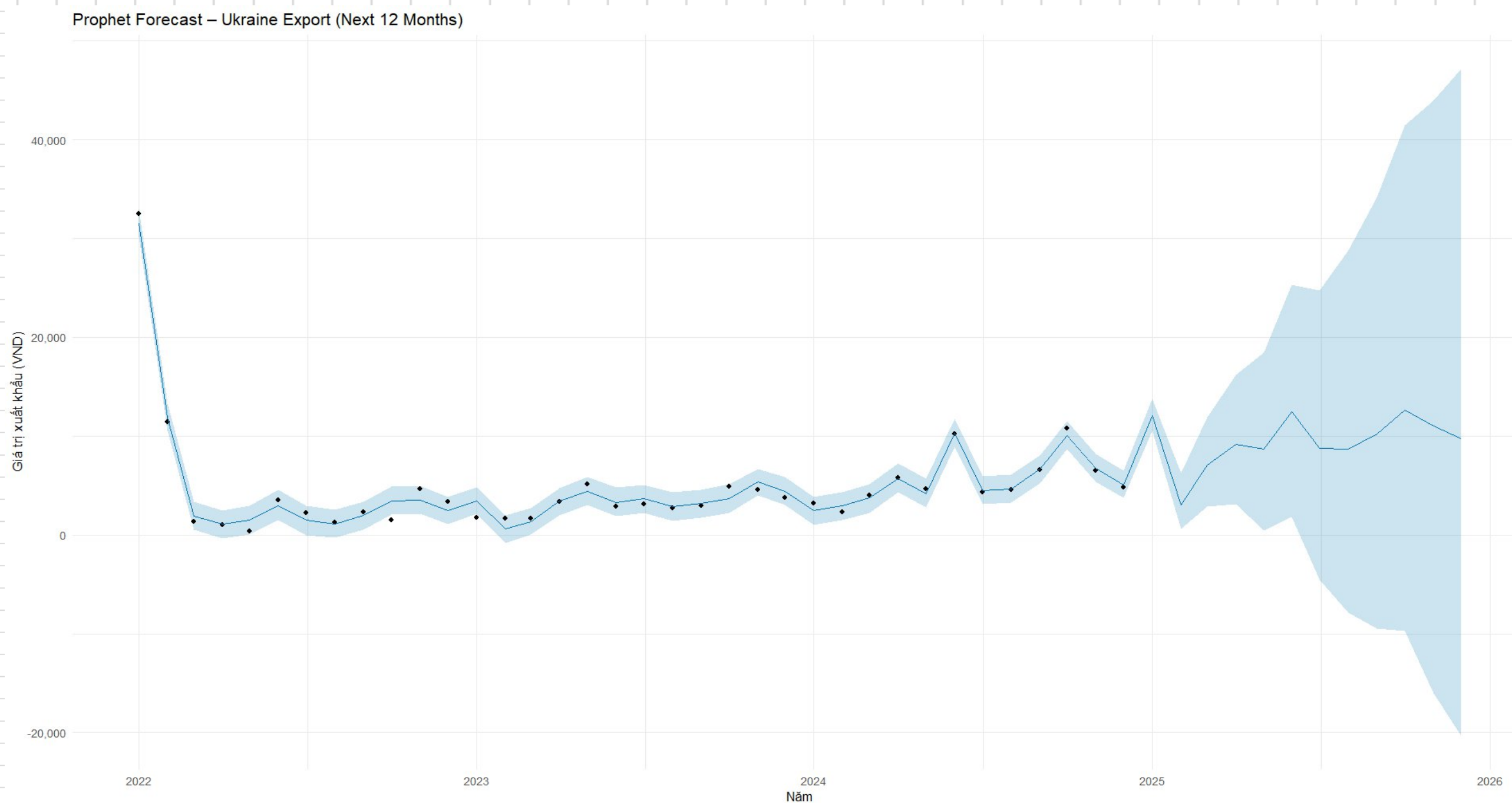
India

So sánh Dự báo Prophet và Dữ liệu Thực tế – India (2025)



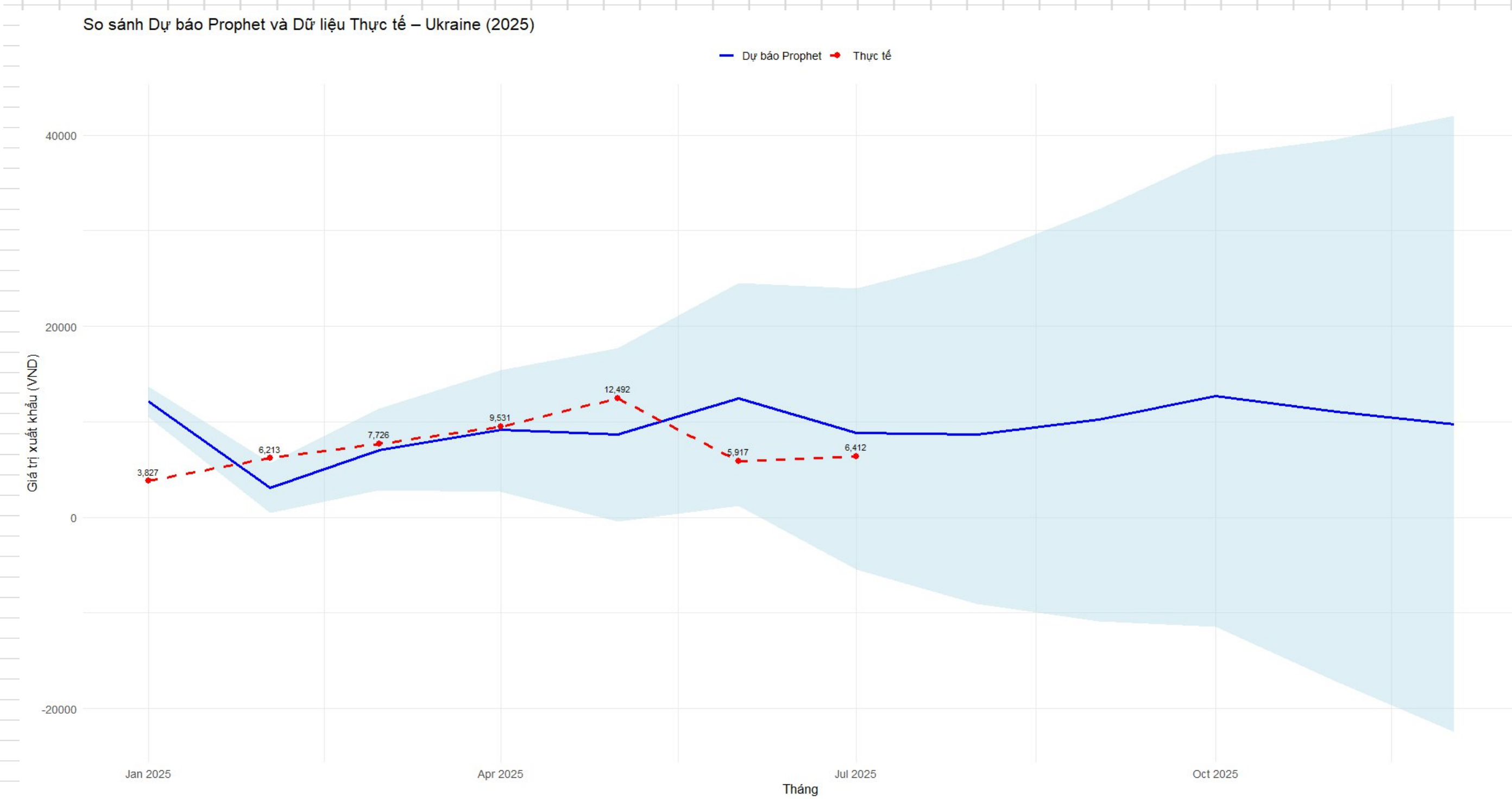
Implement Model

Ukrain
e



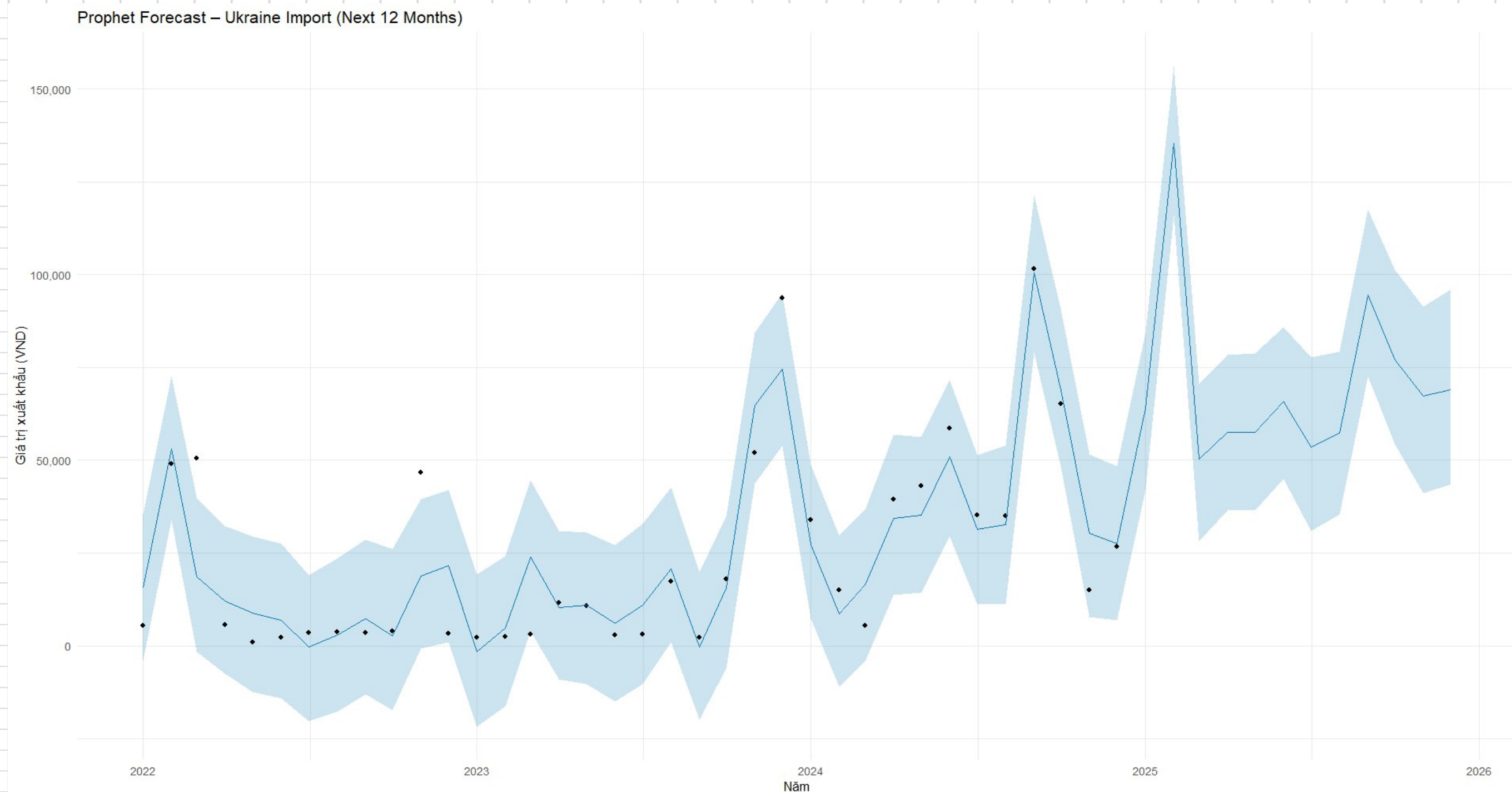
Implement Model

Ukrain
e



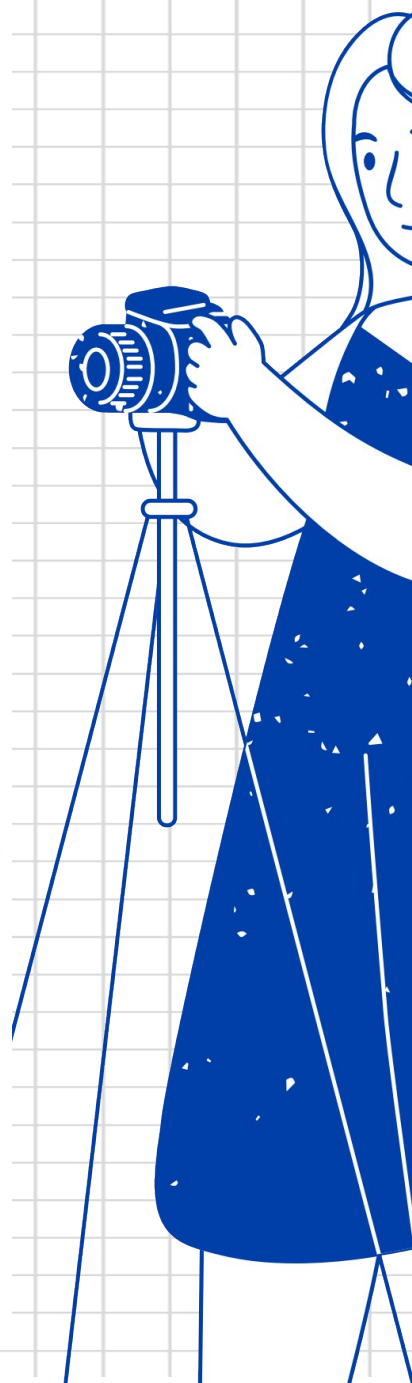
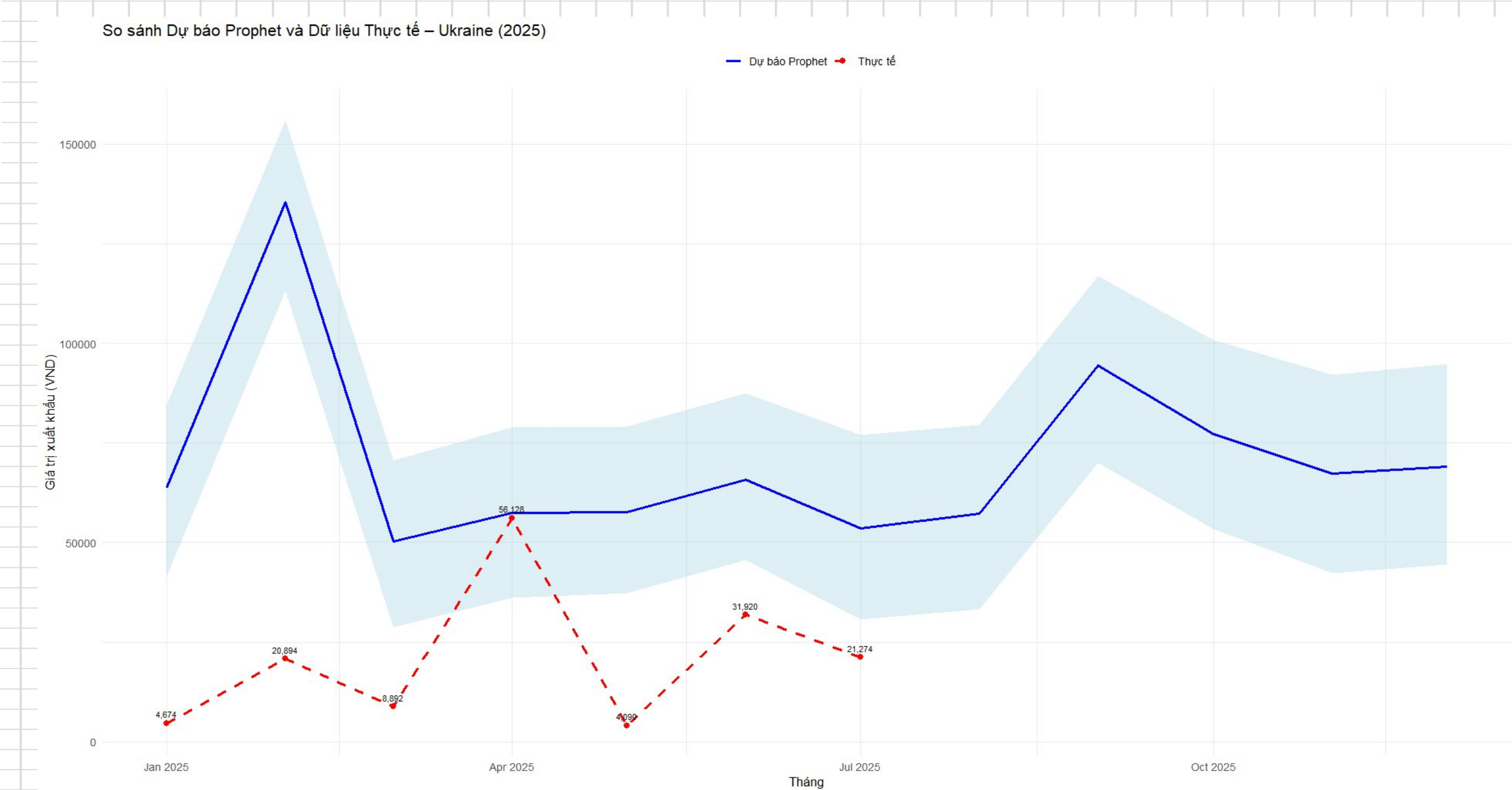
Implement Model

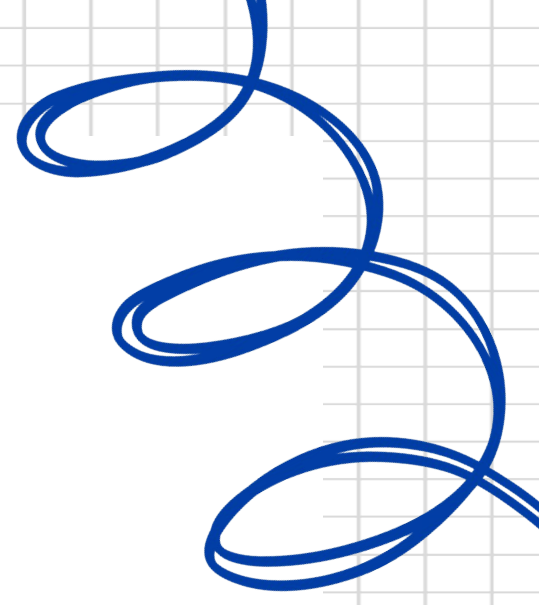
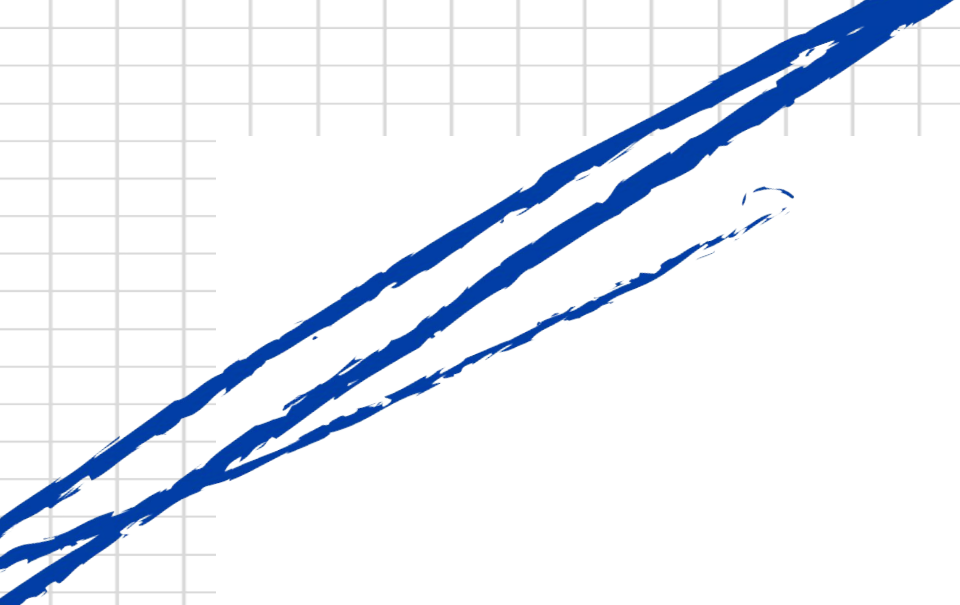
Ukraine



Implement Model

Ukrain
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**THANK
YOU**

